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Milestone 1

SUBMISSION CHECKLIST AND COVER SHEET

DATE: September 17 1997

TO: Dr. Linda Behrens

FROM: TEAM MEMBERS: WASHINGTON, ANGELA
RAGHURAMAN, ANANTH

We are submitting the following Milestone for your evaluation.

[ ] We have selected the following case option (within restrictions specified by the course instructor):

[x] Build your own case option [ ] Running case option

[ ] We are submitting a project definition form.

[x] We are submitting a System Services Form.

[x] We are attaching supporting documentation other than the forms supplied in this casebook.

To be completed by the instructor:

[ ] Milestone returned ungraded because of ________________

____

[ ] Milestone accepted as submitted. Grade of _______ recorded in students' record.

[ ] Milestone accepted as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

[ ] Milestone NOT accepted as submitted. Grade of __________ recorded in students' record.

[ ] Milestone NOT accepted as submitted because ____________

____

resubmit by date of _______________ to establish a permanent grade.

REQUEST FOR SYSTEM SERVICES ATTACHMENT…

The new system will let the students register themselves, from any remote location via a touch-tone phone. The registrar does not have to bother with checking the students' prerequisites or the professors' approval. It is all done automatically.

REQUEST FOR SYSTEM SERVICES

SUBMITTED BY: AUTOMATICA INC. **DATE:** September 10 1997

DEPARTMENT OR **OFFICE:** MAIN OFFICE

TYPE OF REQUEST: [ ] New System - no current computer support

[x] Redesign current computer based system

PROBLEM STATEMENT or **REQUEST DESCRIPTION:**

Charleston Southern University, Charleston, SC, has requested us to automate its manual registration system. In the current system, the registrar register students by manually entering courses into the computer system, while in the new system, students can register from a touch-tone phone. Details are attached.

ADDITIONAL DOCUMENTATION INCLUDED [X] **YES** [ ] **NO**

ACTION (to be completed by the instructor)

[ ] Project approved

[ ] Project approved with suggestions/ variations/ conditions noted below.

[ ] Project rejected

[ ] Decision pending (see instructor and/or comments below)

COMMENTS (to be completed by the instructor)

___

___

___

___

___

MILESTONE 2

SYSTEM PROFILE

MILESTONE 2

SUBMISSION CHECKLIST AND COVER SHEET

DATE: September 23 1997

TO : Dr. Linda Behrens

FROM: TEAM MEMBERS WASHINGTON, ANGELA
RAGHURAMAN, ANANTH

We are submitting the following milestone for your evaluation.

Milestone 2 checklist:

[x] We have completed the system profile for our system.

[x] We are submitting a letter or memorandum of proposal. It has been carefully proofed for spelling and grammatical errors.

[x] We have attached supporting documentation other than the forms supplied in this book. This includes:

- [] Organization Chart
- [x] Sample Forms and Reports
- [] Other______________________________

[] Questions for the instructor:

____

____

____

____

To be completed by the instructor:

I have reviewed Milestone 2. My evaluation is as follows:

[ ]Milestone returned ungraded because of____________________

____

[ ] Milestone acceptable as submitted. Grade of_________ recorded in students' record.

[ ] Milestone acceptable as submitted; however it needs to be corrected and resubmitted to establish a grade.

[ ] Milestone NOT acceptable as submitted. Grade of_______ recorded in students' reocrd.

[ ] Milestone NOT acceptable as submitted because_____________

____

Resubmit by date of_______________ to establish a permanent grade.

AUTOMATICA INC. * 9200 University Blvd. * Charleston SC 29423

September 20, 1997

Mr. Rex Nestor
Head Registrar
Charleston Southern University
9200 University Boulevard

Dear Mr. Nestor:

In the feasibility study we performed, we found your registration process needs to be automated. This would make it more efficient. Allowing students to register themselves without the aid of a registrar representative, would be its main feature. We have two solutions to offer:

1. Students can call from any place if they have access to a touch-tone phone. This solution requires communication software to be added to the existing network.
2. In the second solution, students are allowed to register only from any touch-tone phone on campus. This has the benefit of lower cost since it requires less communication facilities than in the first solution. Although the students have to visit the University in this solution, the pace of the registration remains the same.

We hope to implement one of the above solutions. We're glad to offer our services.

Sincerely,

Ananth Raghuraman

SYSTEM PROFILE

BUSINESS NAME: CHARLESTON SOUTHEN UNIVERSITY

INDUSTRY TYPE: EDUCATION

FUNCTIONAL BUSINESS SYSTEM: THE MANUAL REGISTRATION PROCESS

1) **Problems and opportunities that triggered this project:**

In the current registration process, students go to their respective advisors, with a list of courses they want, to get their (the advisors') approval. Each advisor then goes through the list to see if the student has the necessary prerequisites. If the student doesn't the advisor asks him/her to take another course mix. Then, the student goes to the registrar's office and submits the list. The registrar goes over the list and singles out the courses that have been filled up. These, the student cannot take. The student then has to repeat the process.

Problems with the current system:

A) The student has to go to the registrar's office to register. If it is impossible for the student to do so, for any reason, he/she cannot register.

B) The student has to repeatedly visit his/her advisor if he/she wants to drop/add a course, to get approval for the new addition.

C) The registrar has to manually check for available courses and this is tedious. Even as this process goes on, the other courses may be filled up and the students might end up making another trip to his advisor.

Charleston Southern University has, for the past few years, been making an effort to raise enrollment, and the registration system is one of its business functions it wants to upgrade to provide better service.

Cont...

2) Purpose of the business application, its business goals and the information system objectives directed toward these goals:

Business Application's Purpose: The purpose of the new system is to facilitate the process of registration. It does so by reducing the physical exertion involved in the task of registration.

Business Application's Goals:

- To allow students to register for courses remotely and by themselves, for the most part.
- To reduce the pressure on the advisor, by reducing the number of students who want counseling.
- To reduce the pressure on the registrars.
- To reduce errors, during the process. For example, errors like registering without prerequisites or the registrar registering the student for the wrong course.

Information System's objectives:

- Improved business system performance:
  To make the registration system faster and efficient, so that the student doesn't have to stand in a queue and to update student records as soon as possible so available courses are made known continuously.
- Improved decision making:
  Since the student knows which courses are available, as soon as possible, he/she can immediately decide to add/drop a particular course.
- Lower business costs:
  Since registrars don't have to register the total student population, the university can employ minimum number of personnel; so there is a reduction in total overtime pay.

Cont...

3) The System's users community and their roles:

Direct Users

- o The direct users are students who register for their courses.
- o The other direct users are the registrar representatives
  who register students, who for some reason, come by to register manually.

Managers including: supervisors, middle management, and executive management who will receive information from the system and/or make decision based on information from the system:

- o Rex Nestor, Head Register: He uses the system to prepare detail, exception and summary reports.

Indirect Users

Other departments, office, or divisions which the business who, while not directly using the system, may be affected by the system or receive information it:

- o Indirect user are, the accounting department, business administration office and the Dean of Students.

People or organizations external to the business which will or may affected by the system.

- o Federal and State Financial Aid Departments and other Scholastic/Philanthropic organizations which require a student's registration information to process his/her application for scholarship/finical aid.

4) Capabilities and support the system will provide.

Information system capabilities can be categorized as inputs, ouputs, and decision making opportunities.

Input transactions to be processed:

- o Course addition, During the registration process, each student adds his/her courses.
- o Course drop: A student can drop courses.
- o This Head Registrar can update/view student records.

- o Improved business control:
  This objective aims at better control of university authorities (professors and management etc.) over the registration process. University authorities would be able to check on the status of any student without tedious effort. Only the registrar and authorized personnel would be able to access records. These are the security features.
- o Fewer errors:
  The system aims at reducing human errors during the process.
- o Better service and morale:
  Reducing long queues and errors would make a big impression on the student. Employees would also have more time and energy to devote to other routine tasks.

System policies that constrain the information system solution:

- o The system must obey Federal Privacy Laws, and the Privacy Laws of the university. In simple terms, these laws rule that, without the consent of the student his/her registration information cannot be disclosed to other people.
- o The system shouldn't allow the student to register courses which are taught the same time.

Cont…

- Hold: The Registrar can put a hold on a student's registration.
- Queries for available courses.

Output transactions to be generated:

- Retreival of registration information by students.
- Answering questions about available courses.
- Information retrieval on any student and the facility to update it by the Registrar.

Outputs or Information/Reports to be generated (include a statement of purpose):

Detail Reports:

- Since departments such as Financial Aid want to know all the details of a student's registration, the system generates detailed reports. These reports are also generated as backup, in the event of any mishap involving the hard drive or floppy disks.

Exception reports:

- Exception reports provide for retrieving information on students who have been put on hold, and indicate reason(s).

Summary reports:

- The system provides summary reports so the Head Registrar can report statistical trends in the student population. For e.g., average GPA, minimum GPA and maximum GPA.

Decision support:

- When the student wants to take a course, the system tells him/her if it is available or not.
- If the student, for any reason, wants to register manually, the system tells the registrar whether the student has the required prerequisites.

Cont...

5) All business systems use networks, that is, the distribution structure of people, data, activities, and technology to business locations and the movement of those building blocks between those locations. Remember, each building, each floor, each office may be a different business location. Describe the business network.

Locations where business activity is performed:

- o The registration network consists of the Head Registrar, assistant registrar, representatives, and students. Other people might access the network just to get general information about available courses.
- o Students register for their courses, including dropping, at any remote location, via a touch-tone phone. They dial (803) 863 EXT. and a voice prompt asks them to select from several options (a flow chart of the prototype is enclosed). They can also retrieve other accessible information from that site.
- o Registrars retrieve printed information, transcripts and status checking at the registrar's office.

6) Current technology already supports the other four building blocks of the existing business system. You may design new technology to support various aspects of the new information system later in the project. What hardware and software is being used to capture, store and manage data? What hardware and software is required to support the information system activities described earlier? What communications technology is used to interconnect data and process technology at different business locations described earlier?

Computer Hardware to be used (if known):

A local area network and communications hardware like modems.

Computer software to be used (if known):

An AT&T communication software package will be used.

A Sample Report:

NAME : RUDOLF ENGLEBART

SS# : 990 99 9900

REGISTRATION PIN: 1298UF

SEMESTER : FALL 96

COURSE	CODE	TIME	DAYS	CREDITS	EARNED	GRADE
CS 1	1229845	12:30-2:30	M T R	4	4	A
PSY 101	2292301	12:30-2:30	W F	4	4	A
ENG 122	5322202	9:00-10:30	M T R	3	4	A
C++ PROG	1220102	12:20-2:30	W F	4	4	A

SEMESTER : SPRING 96 (PRESENT)

CS 2	1229923	12:30-2:30	M T R	4
CALC I	1232376	12:30-2:30	W F	4
SAD	1220234	9:00-10:30	M T R	4
DIS MATH	1234578	12:20-2:30	W F	4

STATUS: N

REMARKS: N=NORMAL

H=HOLD

MILESTONE 3

SUBMISSION CHECKLIST AND COVER SHEET

DATE: September 22, 1997

TO : Dr. Linda Behrens

FROM: TEAM MEMBERS WASHINGTON, ANGELA
RAGHURAMAN, ANANTH

[X] We are submitting the following milestone for your evaluation.

Milestone 3 checklist:

[x] We are submitting the following documentation.

- [x] A full letter of proposal (requirement 2)
- [x] A letter describing only the systems development process and management concerns (requirement 1)
- [X] Gantt Chart
- [ ] Other ______________________________

We have reviewed our procedure for the following:

[X] We identified distinct phases within our systems. development approach than can be used to measure the progress of the project.

[X] We described the purpose of each phase in our systems development approach.

[X] We described the inputs used and outputs produced in completing each phase of our system development approach.

[X] We estimated the percent of time required to complete each phase and identified opportunities that exist for working simultaneously on separate phases.

[X] We explained the type and extent of users involvement needed in each phase of our systems development approach.

[X] Our systems development approach requires documentation be generated as a by-product, and several feasibility re-evaluations be made to determine whether the project should be canceled.

Cont…

[X] Our letter was written so that an individual who is not computer literate can thoroughly understand it.

To be completed by the instructor:

I have reviewed Milestone 3. My evaluation follows:

[ ] Milestone returned ungraded because of ________________

____

[ ] Milestone acceptable as submitted. Grade of ________ recorded in students' record.

[ ] Milestone acceptable as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

[ ] Milestone **NOT** acceptable as submitted. Grade of ________ recorded in students record.

[ ] Milestone **NOT** acceptable as submitted because ________

____

Resubmit by date of ______________________ to establish a permanent grade.

AUTOMATICA INC. * 9200 University Blvd. * Charleston,SC 29423

September 20, 1997

Mr. Rex Nestor
Head Registrar
Charleston Southern University
9200 University Boulevard
Charleston, SC 29423

Dear Mr. Nestor:

Automatica Inc. gladly accepts the opportunity to develop faster and efficient registration system. When the system is improved, students will be able to register fastly, and the pressure on the advisors and registrars will be reduced. This will improve student and employee morale, and make a big impression on your customer - the student.

In the following letter, we will explain the procedure that we will used in developing your system. Automatica Inc. looks forward to working for you.

Sincerely,

Angela J. Washington

Angela Washington

AUTOMATICA INC. * 9200 University Blvd. * Charleston,SC 29423

September 20, 1997

Mr. Rex Nestor
Head Registrar
Charleston Southern University
9200 University Boulevard
Charleston, SC 29423

Dear Mr. Nestor:

The following will describe the step-by-step procedures we will follow in developing your registration system. We will divide the project into several phases. On the completion of these phases, your automated registration system will be ready. These phases are:

Survey Phase: The purpose of this phase is to determine whether the problem can be fixed easily or whether it is a complex problem that needs to be solved using the phases listed below. We will meet with you, the assistant registrars, and some students to get an idea of the problem. If the problem is a complex one requiring our efforts, we will establish the project schedule and budget.

Study Phase: The purpose of this phase is to identify and to analyse the problem with the registration system, its causes and effects. In the phase, we will document the system, its strengths, weakness, and uncover new problems. We require your (user/owner) participation as well as the participation of the assistant registrars and students (users) in meetings. As a result, we will be able to frame the objectives and to address the problem.

Definition Phase: The purpose of this phase is to identify what you, the assistant registrars and students want from the new system. We will hold meeting with users, so they can express their needs. we will try to incorporate everyone's demands, so that everybody gets what he/she expects from the system.

Configuration phase: The purpose is to figure out different possible solutions and select one of them. As input, the problem ,as defined in the definition phase, will be used. Each solution will be evaluated with the following criteria: whether the solution is technically feasible, to what degree the solution fulfils user requirements, whether the solution is cost effective, and whether it can be implemented within an acceptable time period. We will figure the technologies for the software and hardware that will be used in developing your automated system. The output from this phase will be the particulars of how the system will be physically designed. This is a key phase that requires your participation and involves other people such as database administrators, software engineers, and programmers. The output from this phase is the software/hardware requirements for the solution.

Procurement phase: The purpose of this phase is to study what software and hardware will best suit the new system's needs and to purchase them. As input, we will use the requirements from the configuration phase. We will research the technology available and the marketplace for software and hardware. Towards this, we will examine the software and hardware available with vendors. At the end of this phase, we will have purchased the necessary software and hardware.

Design phase: The purpose of this phase is to draw up a blueprint for the actual implementation of the system. Obviously, we will need the solution to the problem which we obtain from the configuration phase. We will also need the software and hardware particulars, so we can adjust the design to fit the constraints placed by the available technology. We will be designing the system component by component. We then rapidly develop small working models of each of the components of the system such as database (students' records), security features, and user interfaces. Subsequently, these models will be refined until they all satisfy user expectations. The output of this phase are the technical blueprints (the above models) for construction.

Construction phase: The purpose of this phase is to build the working system. We will use the technical blueprints from the design phase to produce the working systems to construct elements such as databases, applications, and user interfaces. Then, they are tested to see if the system works unison and produce the correct results. The final product is a working system in need of some embellishments.

Delivery phase: The purpose of this phase is to deliver the fully functional registration system. This phase uses the working system from the construction phase. In this phase, we will test to see whether the system works properly. Then, we will convert from the old system to the new one. Activities, such as installing databases prior to conversion, take place in this phase. Finally, the output is a system that is fully functional or a production system as we call it.

System support: Automatica Inc. will provide system support to your new automated system until it is fully functional and running without any technical errors. Automatica Inc. offers maintenance contracts for an additional fee. Our maintenance contracts include fixing software errors that has developed due to user negligence and poor operating of the system environment.

Our system development also includes:

- Collecting information about the new system through meetings and interviews with users and owners, so that we can get feedback on how the system functions. This will enable us to fix current and future problems.
- Documenting information about the system so that other systems analysts/users can learn about system.

The following is with reference to your concerns regarding our system development life cycle:

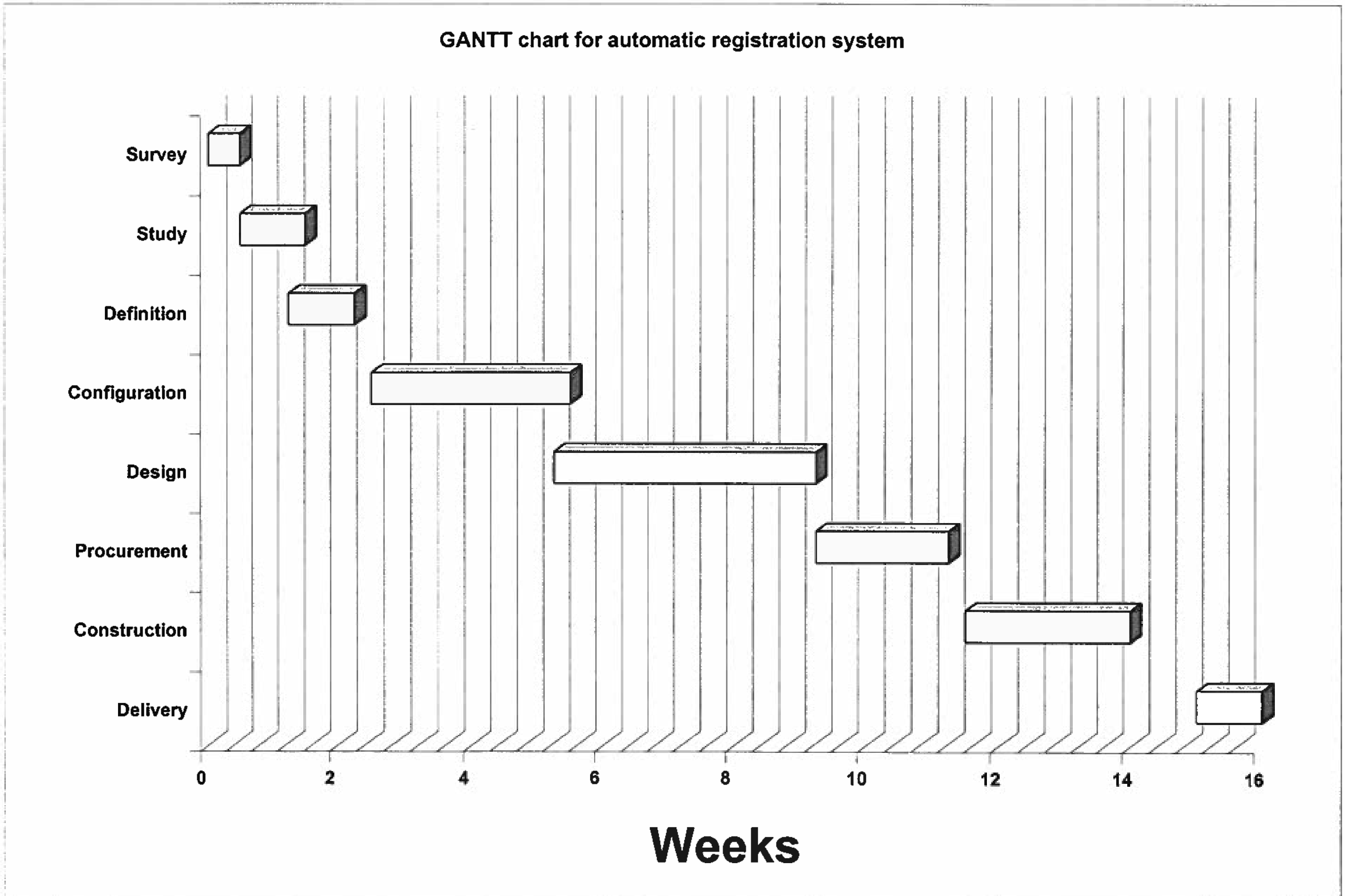
- At the end of each phase, we will include feasibility checkpoints where we will determine if the system we are trying to build is economically feasible, whether it can be completed in an acceptable time period, and whether it can be implemented.
- We will involve the users and owner in every phase so that the final system satisfies your needs.
  In regards to data processing tools, we will communicate with users in as many non-technical term as possible. Pictures and graphical models will help us get the ideas across.

- Recognising that your system is a substantial capital investment, we will investigate all possible cost effective solutions.
- As you wanted, the system development procedures will be divided into phases each of which is associated with a time frame.
- Finally, we assure you that we will use a minimum of computer jargon.

The above is the procedure that we will follow in developing your system to your specification. We look forward to your approval and interest.

Sincerely,

Ananth Raghuraman


GANTT chart for automatic registration system
Survey
Study
Definition
Configuration
Design
Procurement
Construction
Delivery
0
2
4
6
8
10
12
14
16
Weeks

COST ESTIMATES:

AT&T communication software	$2000.00
Hardware	$15000.00
Consultant fees	$25000.00[1]

[1] Includes costs for other systems analysts, installation etc.

MILESTONE 4

SUBMISSION CHECKLIST AND COVER SHEET

Great!

DATE: October 6, 1997

TO : Dr. Linda Behrens

FROM: TEAM MEMBERS WASHINGTON, ANGELA
RAGHURAMAN, ANANTH

We are submitting the following milestone for your evaluation.

Milestone 4 Checklist:

[x] We are resubmitting our System Profile (Milestone 2).

[x] We are submitting an entity relationship diagram for review.

[x] We are submitting a ERD legend and reader's guide.

[x] Our ERD is complemented by a brief written narrative.

[ ] We are submitting the following additional documentation: ____________________

We have carefully checked our entity relationship diagram for the following:

[x] We have used good, descriptive names throughout the entity relationship diagram.

[x] All relationships have been described in both directions.

[x] All inclusive (AND) and exclusive (OR) relationships have been clearly labeled.

[ ] Special comments to the instructor: ____________________

____

____

____

To be completed by the instructor:

I have reviewed Milestone 4. My evaluation follows:

[ ] Milestone returned ungraded because of ____________________

____

[✓] Milestone acceptable as submitted. Grade of 40 recorded in students' record.

[ ] Milestone acceptable as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

[ ] Milestone **NOT** acceptable as submitted. Grade of__________ recorded in students' record.

[ ] Milestone **NOT** acceptable as submitted because____________

____

Resubmit by date of__________________________ to establish a permanent grade.

ERD LEGEND

RELATIONSHIP — A Relationship is a business association between two entities. The name of the relationship is written in the figure. The relationships in the ERD are: <u>Advises</u>, <u>Submits student info</u>, <u>Assigned-to</u>, <u>Requires, Drops</u>, <u>Includes</u>, and <u>Checks for availability</u>.

ENTITY — An Entity is something about which we want to store or capture data. The name of the entity is written in the figure. The Entities in the ERD are: ADVISOR, STUDENT, PREREQUISITE, AVAILABLE COURSE, PRESENTLY REGISTERED COURSE, REGISTRATION PHONE INTERFACE, ADD/DROP PHONE INTERFACE, SCHOLASTIC ORGANIZATION, FINANCIAL AID DEPT., and BUSINESS DEPT. Please note that PREREQUISITE, AVAILABLE COURSE AND PRESENTLY REGISTERED COURSE are different types of COURSE.

Attribute — An Attribute is a characteristic property of an entity. Attributes which are used to identify an entity are called Primary Keys. The names of these are underlined.

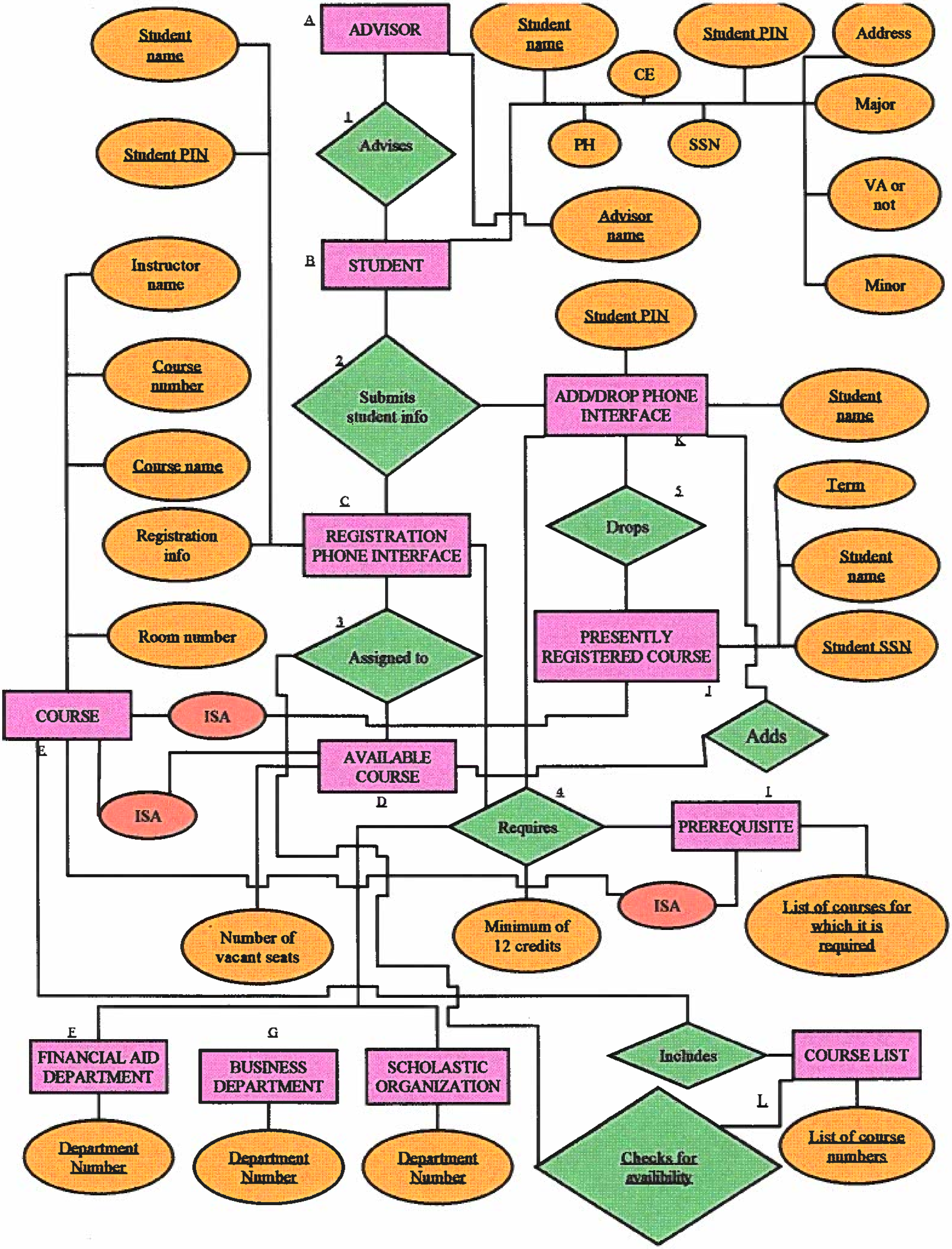
Connections — Which attributes are found which entities, which entities relate to other entities, are all represented using connections or connecting lines.

Cardinality — It's the number of instances for each connecting entity. For example, <u>one</u> ADVISOR can ADVISE <u>many</u> STUDENTS, but <u>a</u> STUDENT can be ADVISED only by <u>one</u> ADVISOR. The representation is given in the following page.

ENTITY	IDENTIFIER	DESCRIPTION
ADVISOR	Advisor name	The ADVISOR advises the student regarding the courses he/she should take. This is after the ADVISOR determines whether the student has taken the prerequisites.
STUDENT	• Student name • Student PIN	He/She is the customer of the University, who is getting an education.
REGISTRATION PHONE INTERFACE	• Student name • Student	It is a feature through which a student can register via a touch-tone phone. All the PIN student has to do is, go to a phone, dial 863-EXT and punch in the course numbers of the courses he/she wants to take and register. It then checks to see if the student has the prerequisites and whether the courses are available.
ADD/DROP PHONE INTERFACE	• Student name • Student PIN	It is similar to the REGISTRATION PHONE INTERFACE, except that it adds a course to the student's course list or drops a course. It requires the student to be registered already, for the current semester.

ENTITY	IDENTIFIER	DESCRIPTION
PRESENTLY REGISTERED COURSE	• Course name • Course number • Term	It is the subject the student has registered for in the current semester
AVAILABLE COURSE	• Course name • Course number • Number of vacant	It is a course which has vacant seats, and for which seats students can register.
PREREQUISITE	• Course name • Course number • List of course numbers for which it is required.	A course which is required to be taken before another one can be.
COURSE LIST	• List of course course names and numbers	It is a list of courses which the University offers.

ENTITY	**IDENTIFIER**	**DESCRIPTION**
FINANCIAL AID	Dept. Number	It is a department the university which processes the financial aid needs of students. For this, it requires the registration information of students.
BUSINESS DEPT	Dept. number	It charges student according to the number of credits they take in the semester. For this, it needs the registration information of students.
SCHOLASTIC	Organization name	An Organization to which the student applies for aid.



A
ADVISOR
Student name
Student PIN
Address
CE
Major
PH
SSN
1
Advises
Student name
Student PIN
VA or not
Advisor name
B
STUDENT
Minor
Instructor name
Student PIN
2
Submits student info
ADD/DROP PHONE INTERFACE
Student name
Course number
K
Course name
5
Term
C
Drops
Registration info
REGISTRATION PHONE INTERFACE
Student name
Room number
3
Assigned to
PRESENTLY REGISTERED COURSE
Student SSN
J
COURSE
ISA
Adds
E
AVAILABLE COURSE
ISA
D
4
Requires
I
PREREQUISITE
ISA
List of courses for which it is required
Number of vacant seats
Minimum of 12 credits
F
G
FINANCIAL AID DEPARTMENT
BUSINESS DEPARTMENT
SCHOLASTIC ORGANIZATION
Includes
COURSE LIST
L
Department Number
Department Number
Department Number
Checks for availibility
List of course numbers

Reader's guide

In the ERD, the Registration process starts with the entity ADVISOR (marked A). The ADVISOR Advises(1) STUDENTS(B). One ADVISOR may be assigned to many students, while a STUDENT can have only one advisor. A STUDENT registers via a touch-tone phone, which is connected to a network of computers which contain his/her records. This is the REGISTRATION PHONE INTERFACE(C). A student submits student info(2) throughout entity. A REGISTRATION PHONE INTERFACE corresponds to exactly one STUDENT. The REGISTRATION PHONE INTERFACE (which in the current system is the REGISTRATION FORM) registers the student. In order to do this, it Requires(3) the student to have taken the PREREQUISITES(I). The AVAILABLE COURSE(D) is then assigned to the student. Available because the system will not assign a course which has been filled up. The AVAILABLE COURSE now becomes a PRESENTLY REGISTERED COURSE(J), which is defined in the legend. If a student wants to add or drop a course, he encounters the ADD/DROP PHONE INTERFACE(K). To Add(8) a course, the student enters the course number and the system checks for availability(6) with the COURSE LIST(L). If a student wants to Drop a course, the ADD/DROP PHONE INTERFACE Requires(4) that the student have a minimum of 12 credits after doing so.

There are several external users of the registration system who need student's registration information. They are, the FINANCIAL AID DEPT.(F), BUSINESS DEPT.(G) and any SCHOLASTIC ORGANIZATION to which the student may have applied for aid. All of them Require that the student have a minimum of 12 credits (an attribute of Requires).

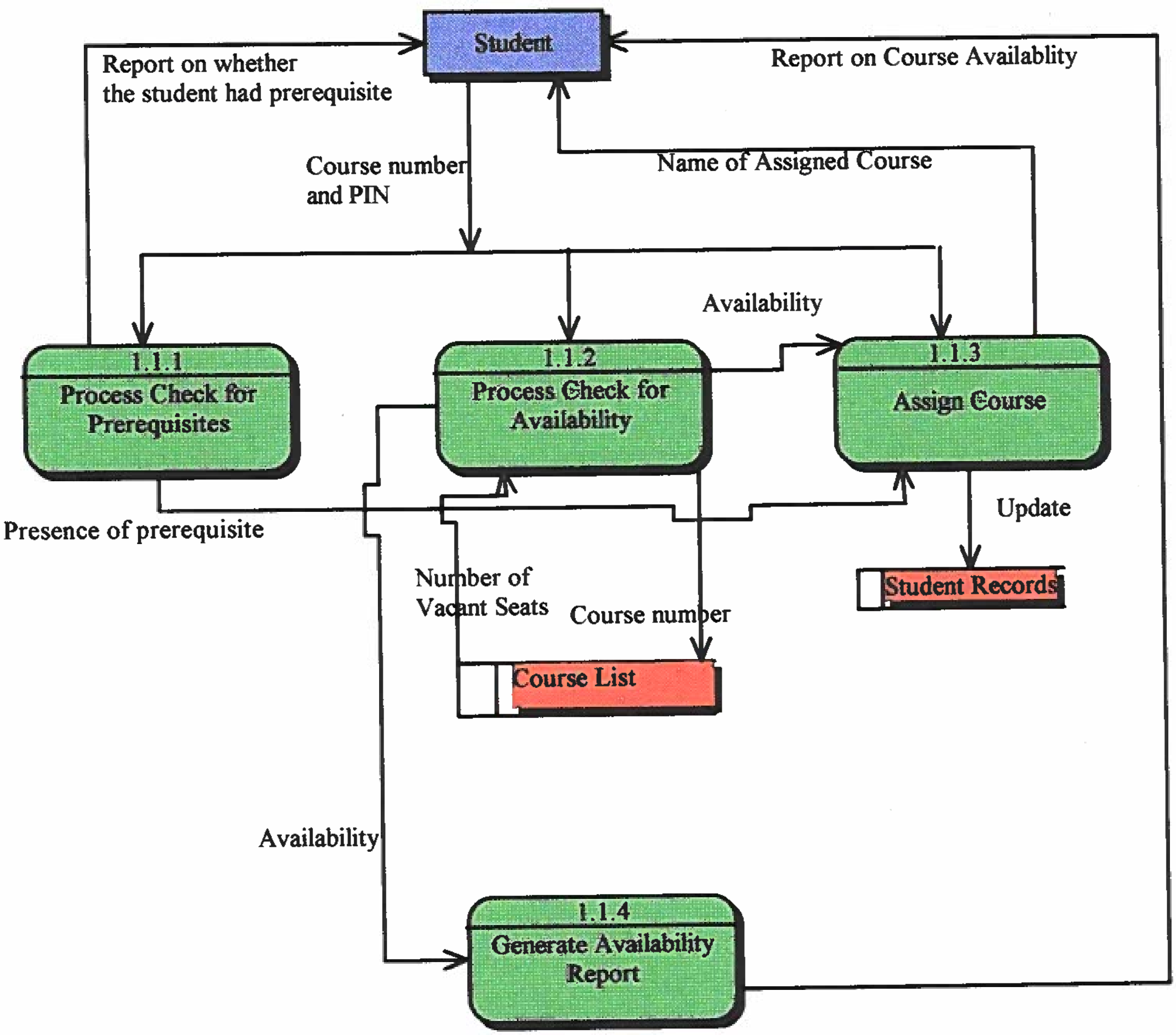
CARDINALITIES:

- An ADVISOR can be assigned to many students, but a STUDENT can be assigned to only one ADVISOR.
- A STUDENT corresponds to exactly one REGISTRATION PHONE INTERFACE and vice-versa.

- A STUDENT corresponds to exactly one ADD/DROP PHONE INTERFACE and vice-versa.
- A REGISTRATION PHONE INTERFACE is assigned to one or many AVAILABLE COURSEs and vice-versa.
- A REGISTRATION PHONE INTERFACE might require one or more PREREQUISITEs, while a PREREQUISITE might require zero or more REGISTRATION PHONE INTERFACEs (each actually corresponds to a student).
- Zero or many REGISTRATION PHONE INTERFACES are required by FINANCIAL AID DEPT., BUSINESS ORGANIZATION, and SCHOLASTIC ORGANIZATION, while a REGISTRATION PHONE INTERFACE requires one FINANCIAL AID DEPT., BUSINESS DEPT., and zero or many SCHOLASTIC ORGANIZATION.
- The ADD/DROP PHONE INTERFACE can replace the REGISTRATION PHONE INTERFACE in all the above relationships.
- In addition, the ADD/DROP PHONE INTERFACE <u>Drops</u> one or many PRESENTLY REGISTERED COURSES (making sure that there are at least 12 credits after doing so). A PRESENTLY REGISTERED COURSE can be dropped by zero or many ADD/DROP PHONE INTERFACEs.
- The ADD/DROP PHONE INTERFACE also <u>Adds</u> courses. It can add one or many AVAILABLE COURSE. An AVAILABLE COURSE can be added by zero or many ADD/DROP PHONE INTERFACES.
- An ADD/DROP PHONE INTERFACE <u>checks for availability</u> (of the course the student wants to add)with one COURSE LIST. A COURSE LIST can be checked for availability of courses by

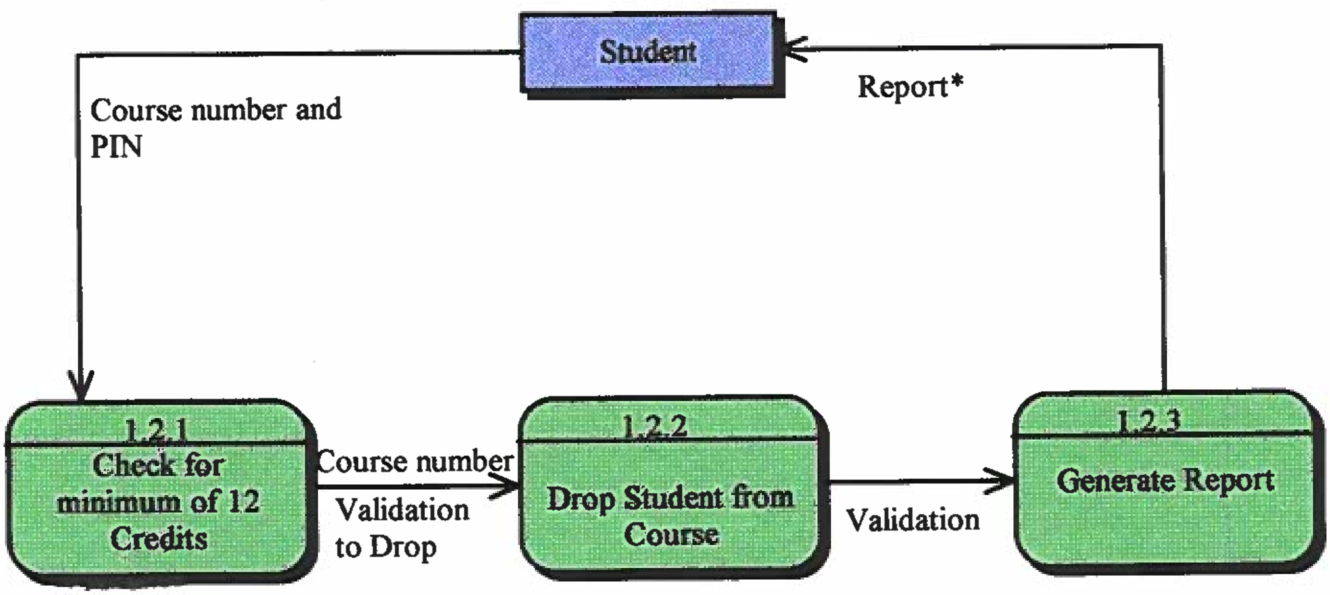
- PREREQUISITE, AVAILABLE COURSE, and PRESENTLY REGISTERED COURSE are all COURSEs. They inherit the attributes, Course name and Course number. Likewise, the attributes of the three are inherited by COURSE.

DFD for Course Addition Subsystem
Level 2
Explosion of 1.1

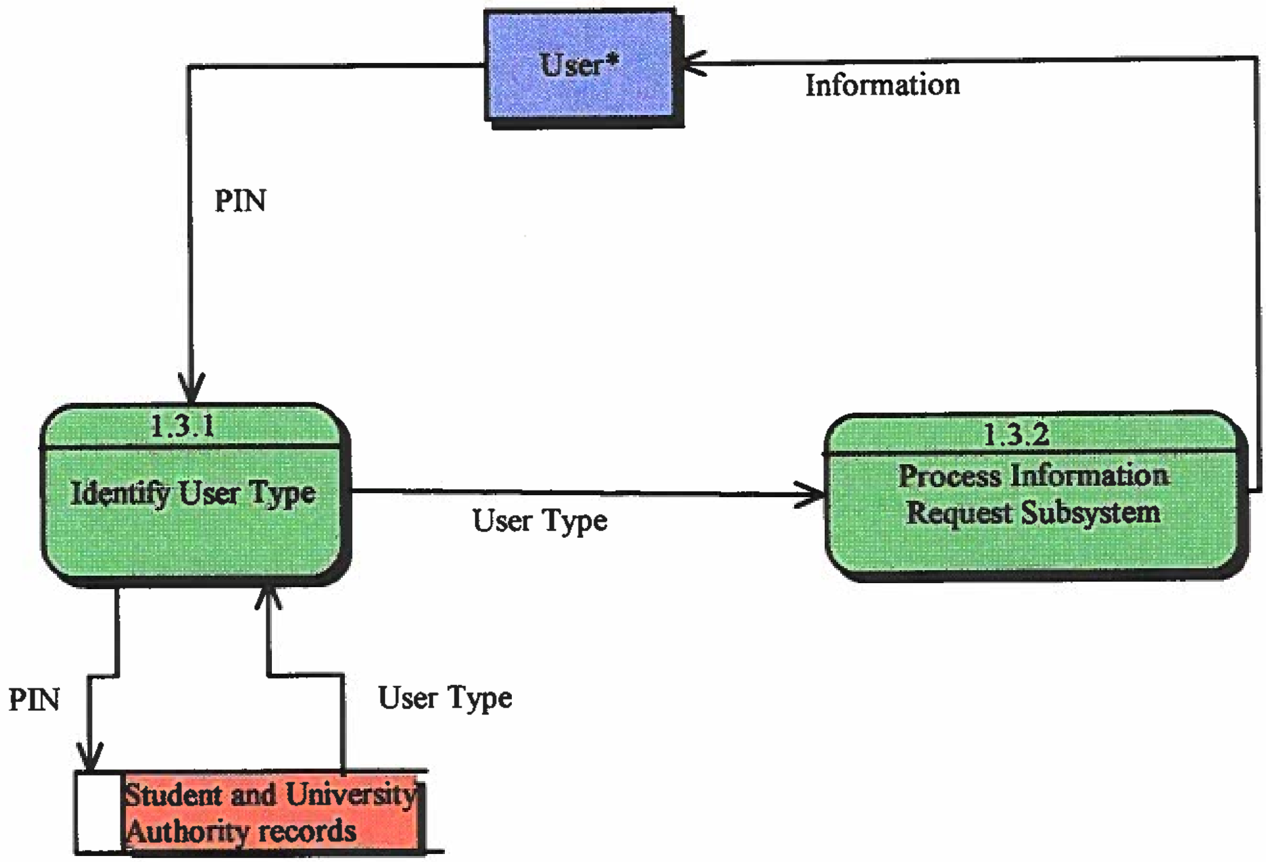


Go to page 15 for explosion of 1.1.1

DFD for Course Drop Subsystem
Level 2 (explosion of 1.2)

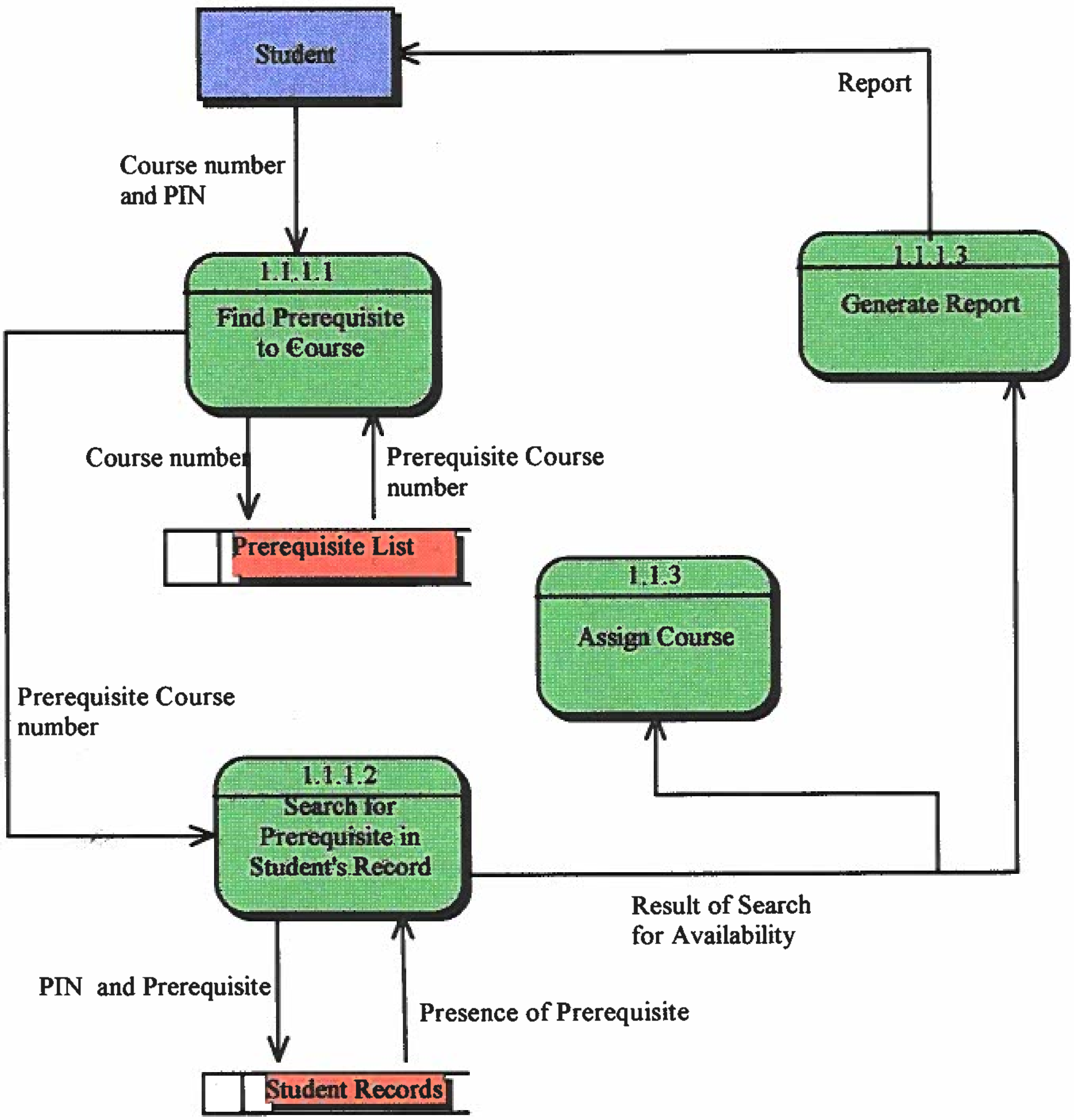


DFD for Information Retrieval Subsystem
Level 2
Explosion of 1.3

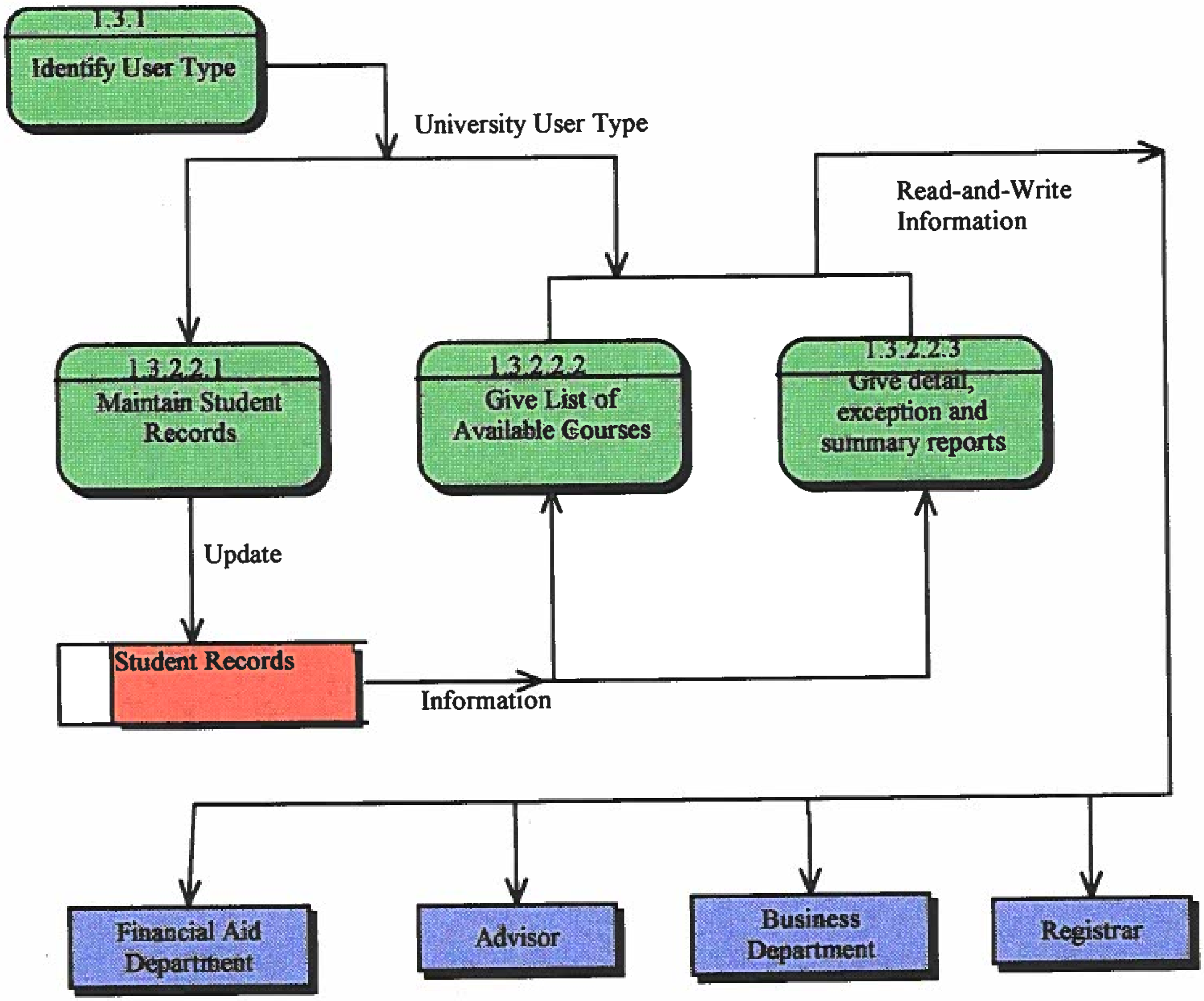


Go to page 17 for explosion of 1.3.2

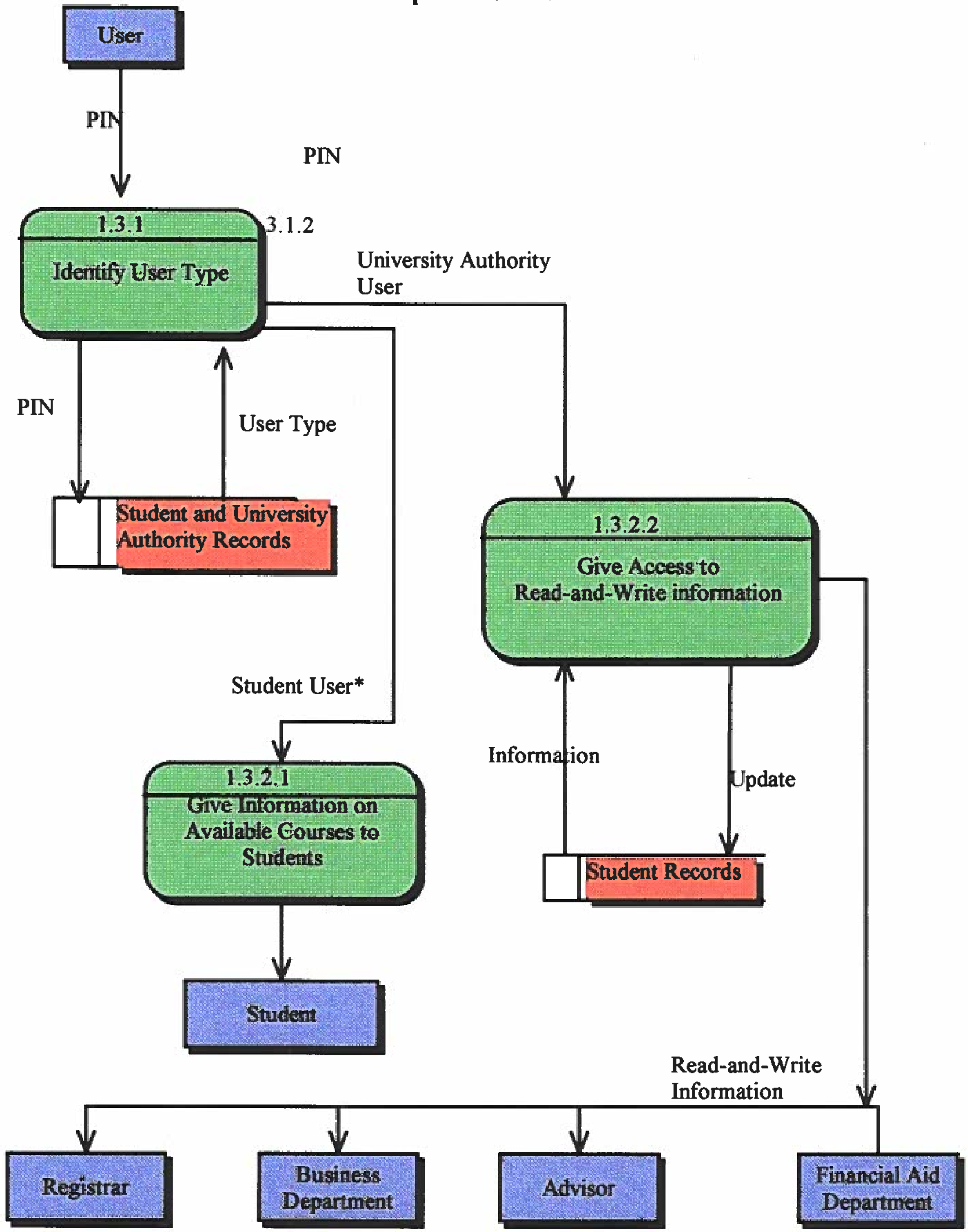
DFD for Course Addition Subsystem
Level 3
Explosion of 1.1.1



DFD for Information Retrieval Subsystem
Level 4
Explosion of 1.3.2.2



DFD for Information Retrieval Subsystem
Level 3
Explosion of 1.3.2



Go to page 16 for explosion of 1.3.2.2

A brief narrative of the decomposition diagram

The system we are modeling is the **Registration system**. It is made up of three systems: Course Addition Subsystem which adds courses, Course Drop Subsystem, which drops courses, Information Retrieval/Update Subsystem which gives information on the list of available courses, by updating student records by University authorities and retrieval of detail, exception and summary reports.

The **Course Addition Subsystem** which adds courses, consists of PROCESS PREREQUISITE CHECK, PROCESS AVAILABILITY CHECK and ASSIGN COURSE. PROCESS PREREQUISITE CHECK checks whether the student has the necessary prerequisite to add a particular course. PROCESS AVAILABILITY CHECK checks whether the course the student wants to take is available. ASSIGN COURSE assigns the course after the above two processes are successful.

The **Course Drop Subsystem** which drops courses, consists of PROCESS CHECK FOR MINIMUM OF 12 CREDITS, PROCESS COURSE DROP. PROCESS checks whether the student would have a minimum of 12 credits after dropping the course. PROCESS COURSE DROP drops the course and updates the student's record.

The **Information Retrieval Subsystem** consists of IDENTIFICATION OF USER TYPE. User Type means whether the user is a student or university authority (Advisor, Head Registrar, Assistant registrars, Business Department and Financial Aid Department). PROCESS INFORMATION REQUEST processes requests for information by students and university authorities. For students, it gives read-only access to the list of courses available and read-and-write access to university authorities. In addition, it maintains students records and provides detail, exception and summary reports.

Cont...

The **Security Check Subsystem** allows only users who are authorized by the university, to access information. This is made possible by the use of a **PIN**. This enables selective access and updating of information.

Brief narrative of Level 2

In the **"explosion" of Course Addition Subsystem**, the main external agent is the student; the student enters his/her PIN and the course number(s) he wants to add. The system checks whether the student has the prerequisites (refer to brief narrative of Level 3) and checks whether it is available.It does this by finding out the number vacant seats in the course. This information is then used to generate a report in order to inform the student whether it is available. This data is also used to determine whether the course can be assigned. When the course is assigned, the student's record is automatically updated (through updating the Data Store Student Records) . The student is then given the names of the courses to which he/she is assigned.

In the "**explosion of Course Drop Subsystem (1.2)**", the external agent is the Student. He/she inputs his/her Course number(s) which he/she wants to drop and the PIN. The system then checks whether the student would have a minimum of 12 credits after dropping. The course number and validation (whether or not the course(s) should be dropped) is the data for the next process Drop Student from Course. Depending on the validation, it drops the student or not. If it does, it updates the student's records. The validation is then used to generate a report; this report then informs the student if the course(s) has been dropped.

Information Retrieval Subsystem has been exploded to depict the following processes: A user inputs their PIN and the Process Identification of User Type identifies the user as student or university authority. If the user is a student, he/she is given access only to the list of available courses (this is read-only). If the user is a university authority, then the process (Give Read-and-Write Access to All Student Records) provides for retrieval as well as updating of information.

Brief narrative of Level 1

This level of Data Flow Diagrams id formed by depicting the general data flows between the most important External agents and the systems. As we can see, the systems are given names 1.1, 1.2, 1.3 and 1.4. The important external agent in 1.1, 1.2 and 1.3 is the STUDENT. The student inputs his/her PIN and the course number(s) of the course(s) he/she wants to add or drop. Please note that registering for the first time is the same as adding a course. In 1.1 (Course Addition Subsystem) and 1.2 (Course Drop Subsystem), the records of the student, which are part of student records (Data Store), are updated.

In 1.3 (Information Retrieval Subsystem), the student inputs the PIN and gets the list of courses which are available at the time. The business department, Financial Aid department, Registrars and Advisors can retrieve information as well as update it. For this, they have to first input their PIN. The arrows going from them to "Information Retrieval Subsystem" show this.

A brief narrative of Level 3

This level is an "**explosion of Process Check for Prerequisites**". The student inputs his/her course number and PIN. The process then finds the course corresponding to the course (FIND PREREQUISITE CORRESPONDING TO COURSE) using the course number and PREREQUISITE LIST. This gives the course number of the prerequisite. This course number of the prerequisite is then used to check whether the student had taken the course using his records in Student Records (SEARCH FOR PREREQUISITE IN STUDENT'S RECORD).

The result (whether the prerequisite was found or not) is then used in the Process Assign Course. It is also used to give a report to the student as to the successfulness of his/her attempt.

Reader’s guide

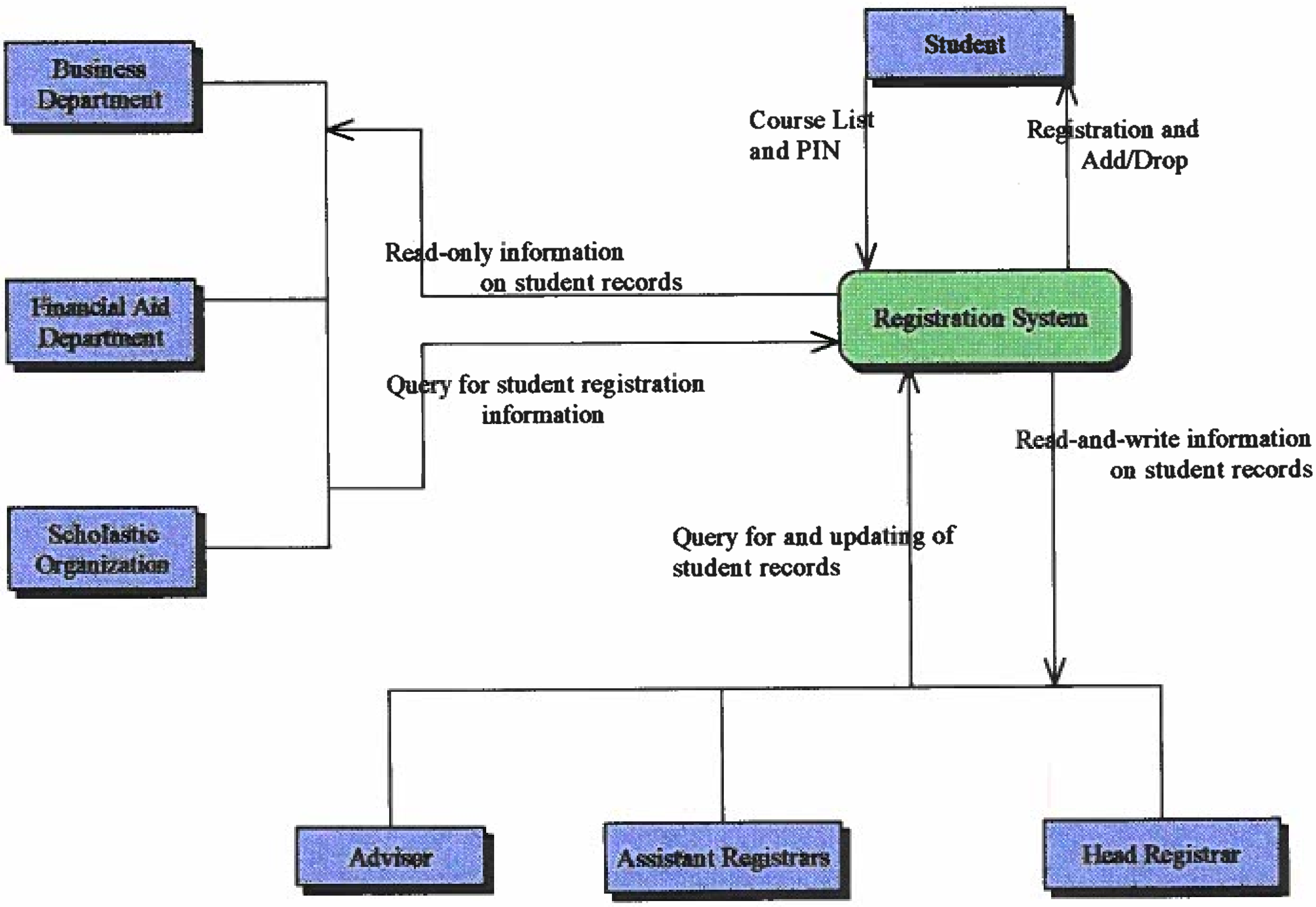
The process of registration starts with the student adding courses. This is done by the Course Addition Subsystem. Another function is dropping courses. This is done by the Course Drop Subsystem. Finally, the departments (Business and Financial Aid), Advisors, Registrars and other Scholastic Organizations may want to retrieve or update students’ registration information. This function is done by the Information Retrieval Subsystem.

To first enter the system, the user must enter his PIN or registration identification number which would be assigned to him/her. The Security Check Subsystem then compares this PIN with its database of records on students and university authorities. If it finds a match, it allows the user to access the system. If the user is a student and wants to register, he/she enters the course numbers of the courses he/she wants to register. The Course Addition Subsystem then checks to see if the student has the necessary prerequisites. After this, the subsystem then checks to see if the course hasn’t been filled up. If at least one of these happens, it tells the student that he/she cannot register for the course and doesn’t assign the course. If both of these are not true, it assigns the course to the student. It then repeats the processes for the next course number. If the student wants to drop a course, the Course Drop Subsystem takes over. Again the student has to enter the course number(s) of the courses he wants to drop. The system then checks to see if the student would have a minimum of 12 credits after the drop. If he would, it doesn't allow him to drop. Otherwise, it allows him/her to do so. It then updates the record of the student in the database. The it reports to the student about the success of the student's attempt.

The information Retrieval Subsystem is there so that users can retrieve or update information. Users can be classified into two groups: students and university authorities. Students can access the subsystem only to get list of available courses whereas university authorities (Business and Financial Aid departments, Advisors and Registrars) can retrieve as well as change a student's registration information in the database. When the user enters the subsystem, the system first identifies what type of user he/she is - whether a student or university authority. After this, if the user is a university authority, it gives him/her access to change information. It also gives him detail, summary and exception reports.

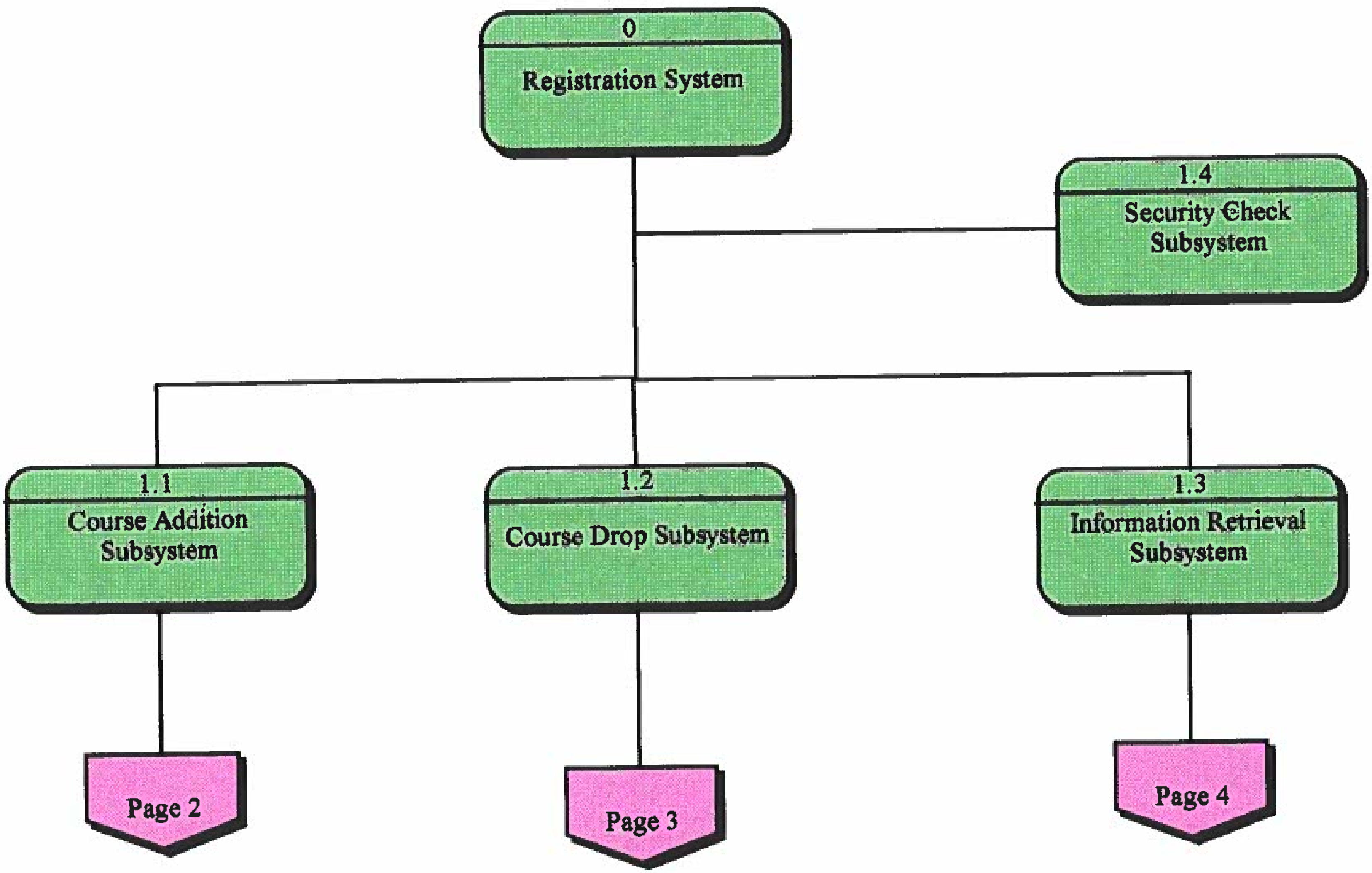
Finally, the **Scholastic Organizations** want a student's registration information because students have applied for financial aid. they cannot retrieve information directly because they are external organizations. So they have to retrieve the information indirectly through the Registrar's Office.

CONTEXT DIAGRAM FOR CSU's REGISTRATION SYSTEM



FUNCTIONAL DECOMPOSITION DIAGRAM OF CSU's REGISTRATION SYSTEM

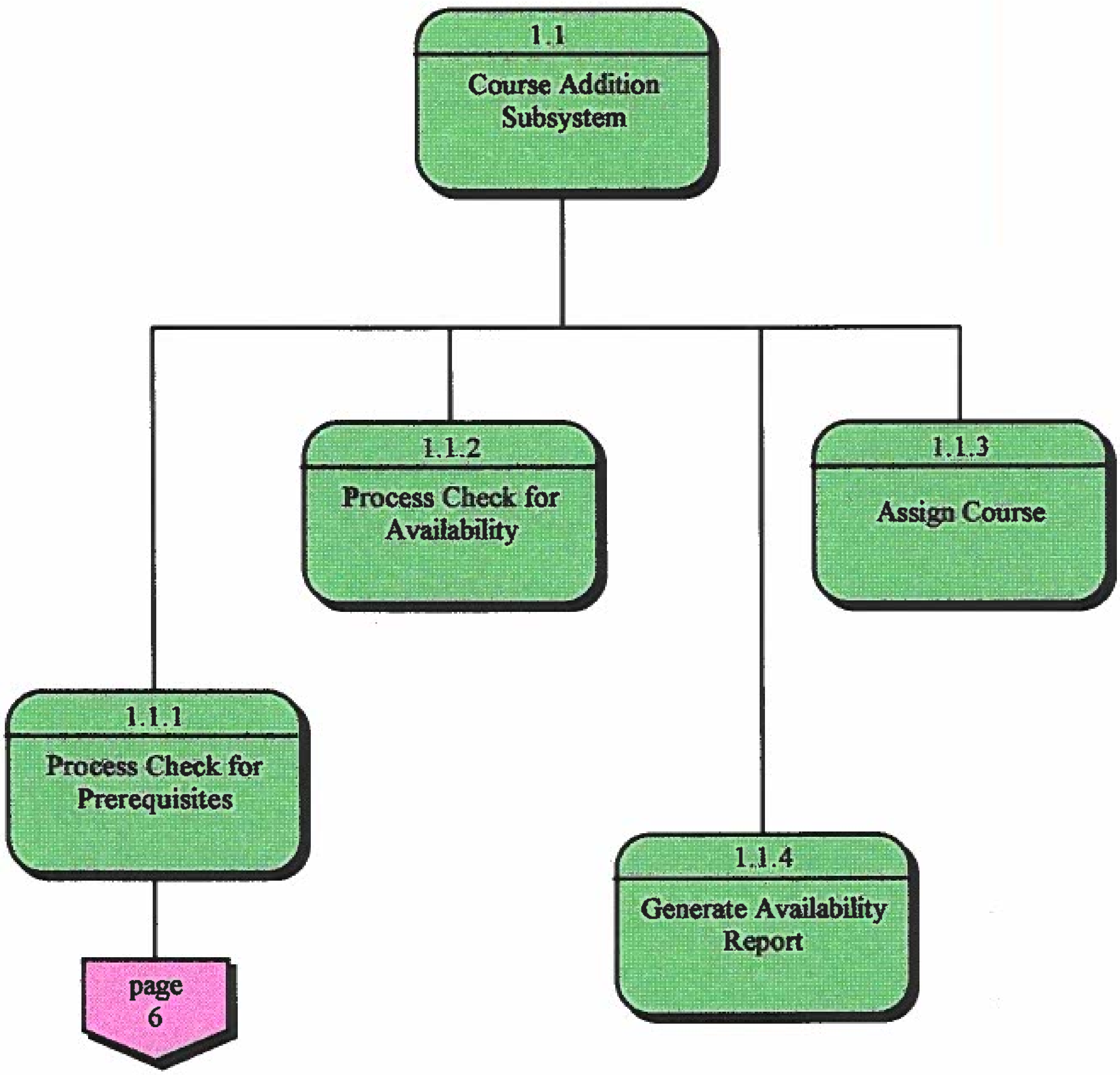
LEVEL 1



DECOMPOSITION DIAGRAM cont...

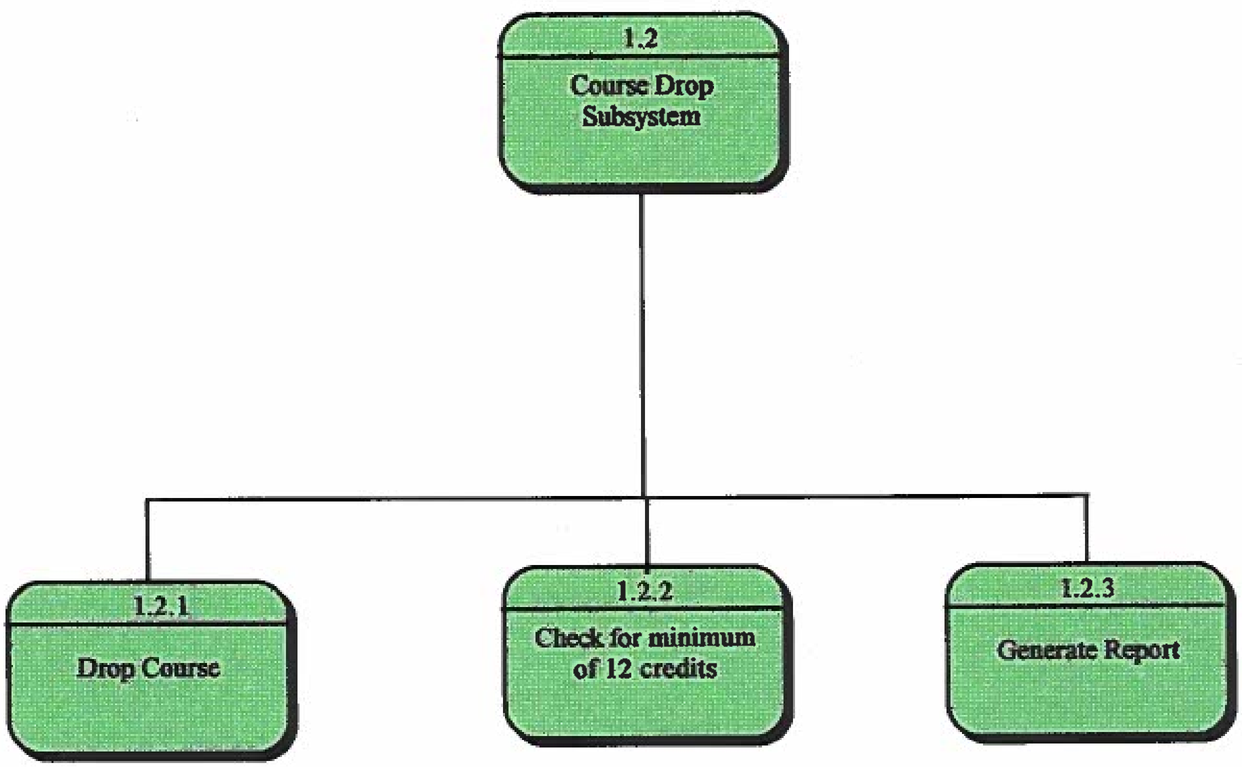
LEVEL 2

EXPLOSION OF 1.1



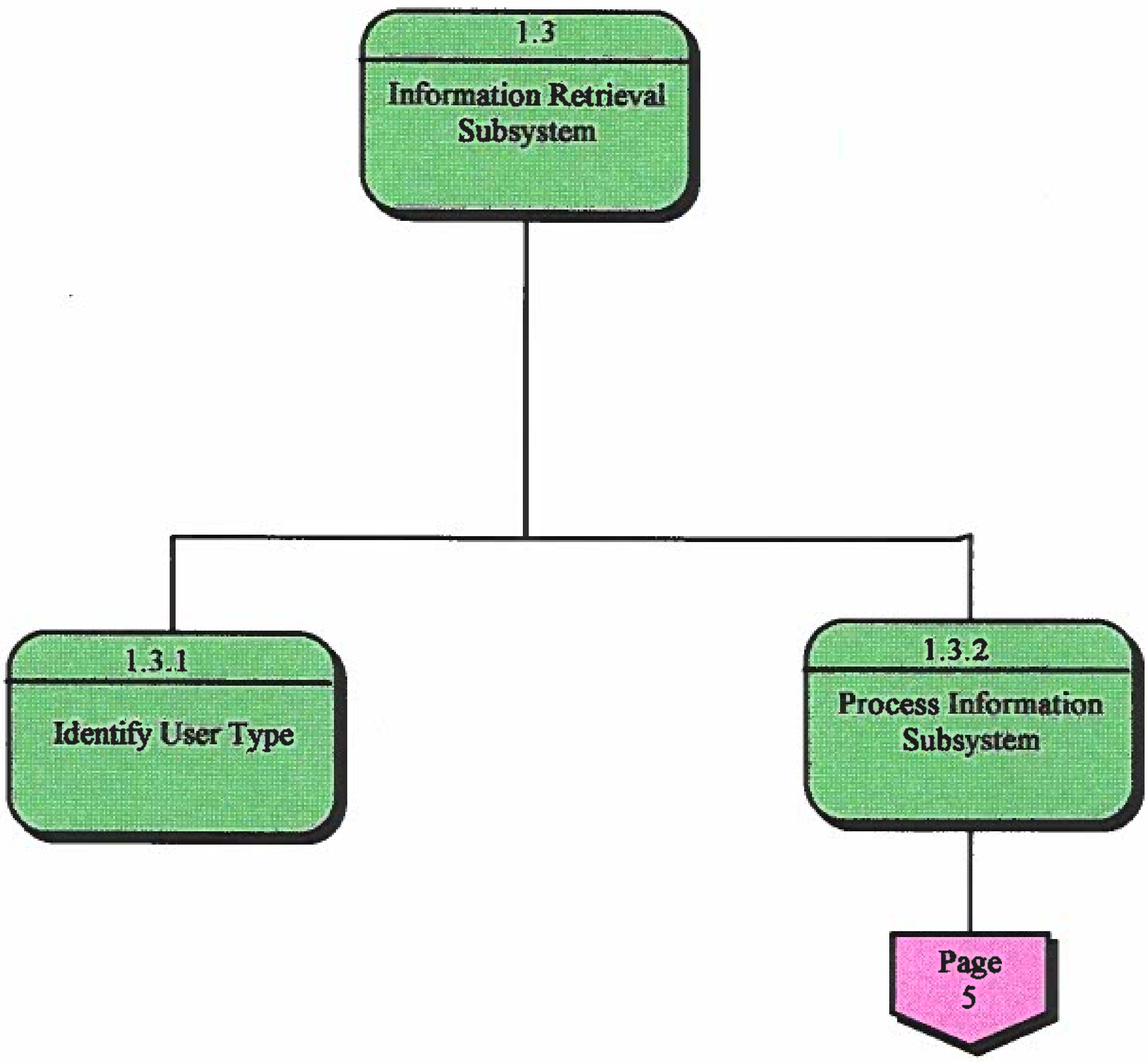
DECOMPOSITION DIAGRAM cont...

LEVEL 2
EXPLOSION OF 1.2

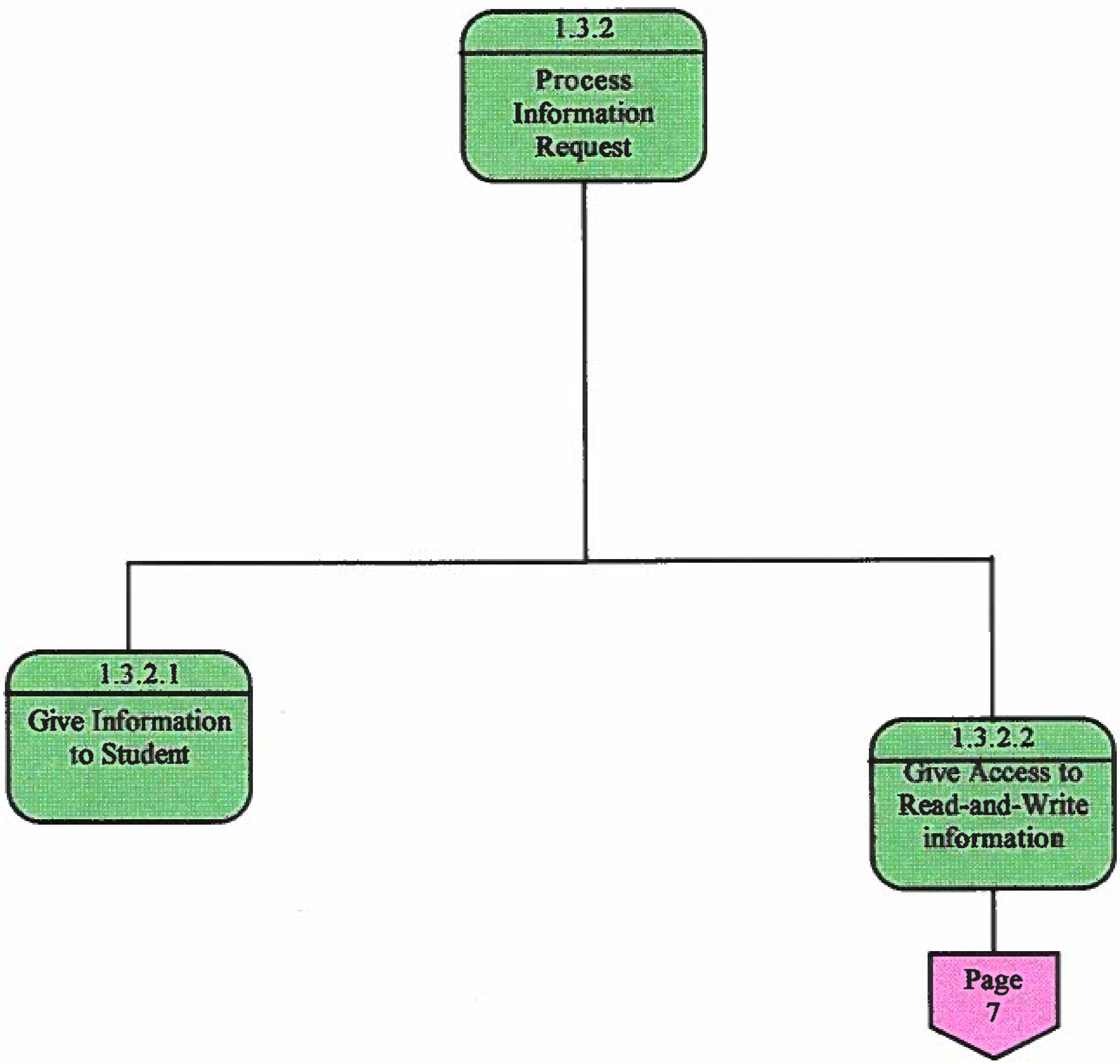


DECOMPOSITION DIAGRAM cont...

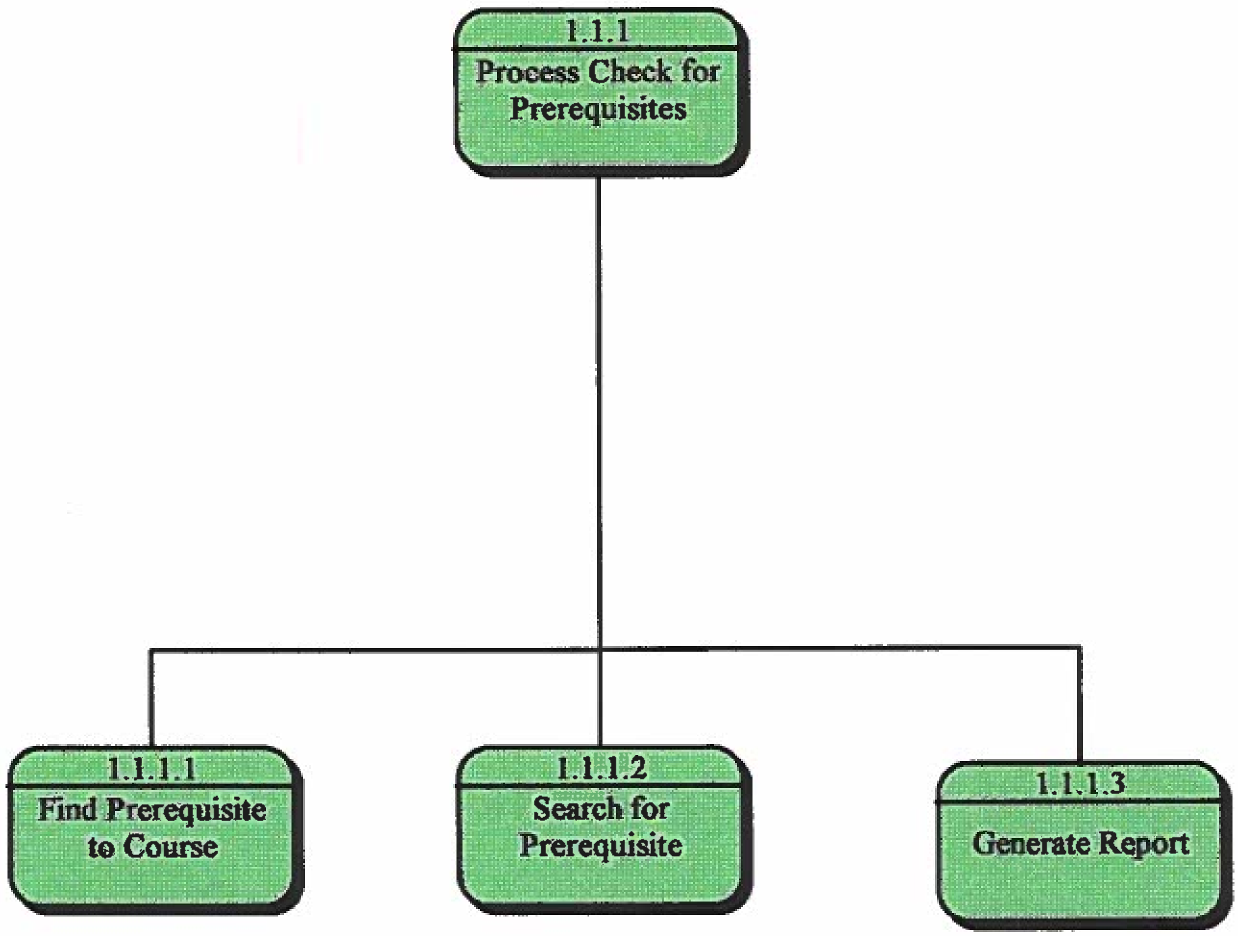
LEVEL 2
EXPLOSION OF 1.3



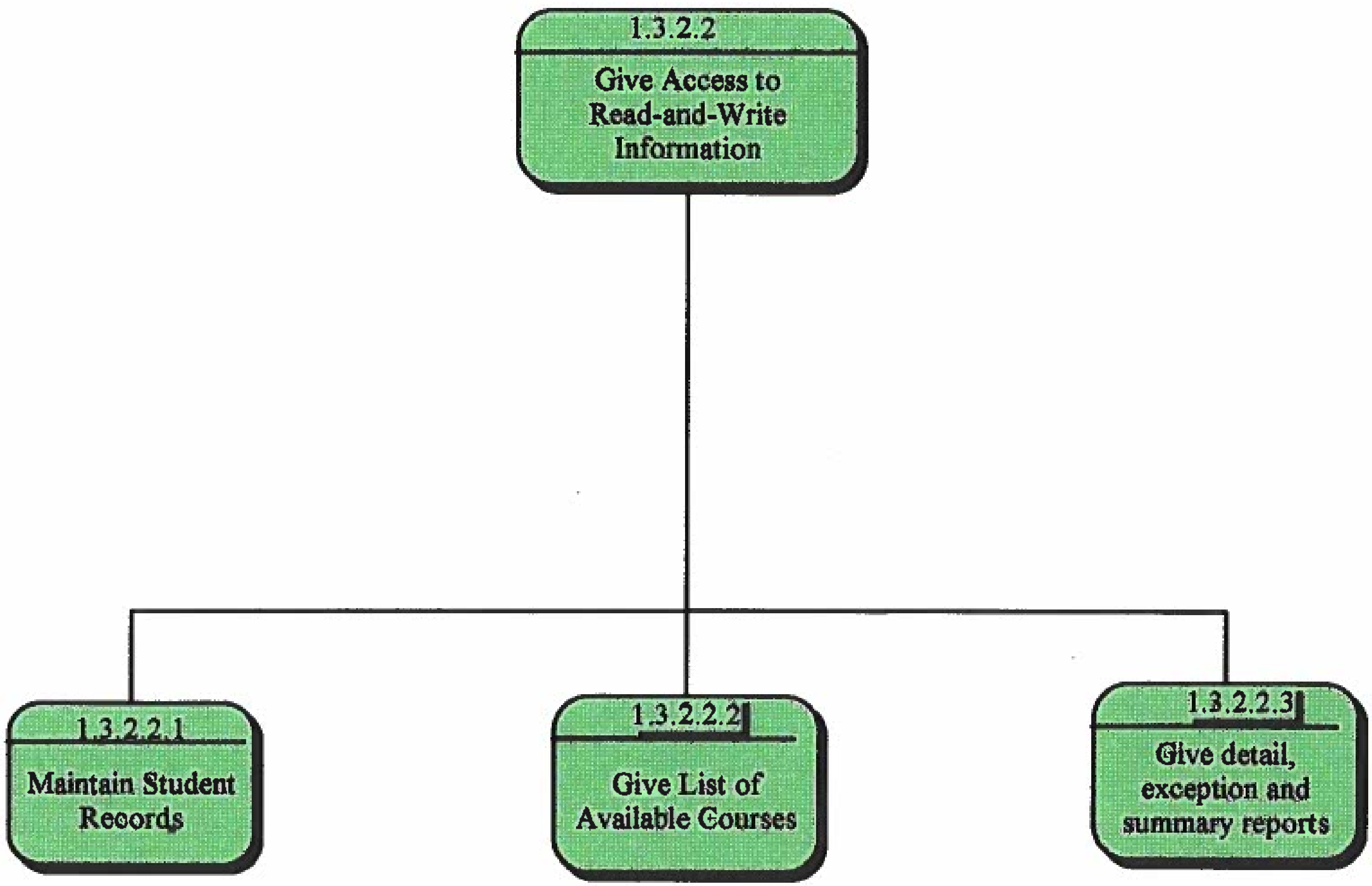
DECOMPOSITION DIAGRAM cont...
LEVEL 3
EXPLOSION OF 1.3.2



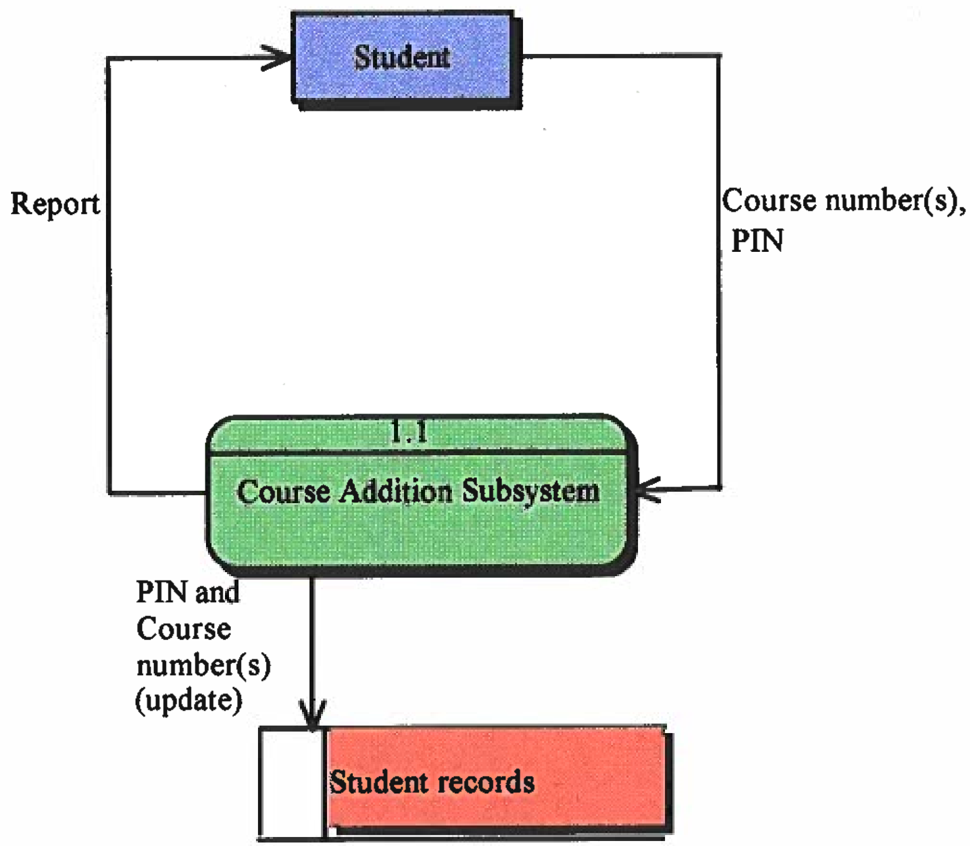
DECOMPOSITION DIAGRAM
LEVEL 3
EXPLOSION OF 1.1.1



DECOMPOSITION DIAGRAM
LEVEL 4
EXPLOSION OF 1.3.2.2

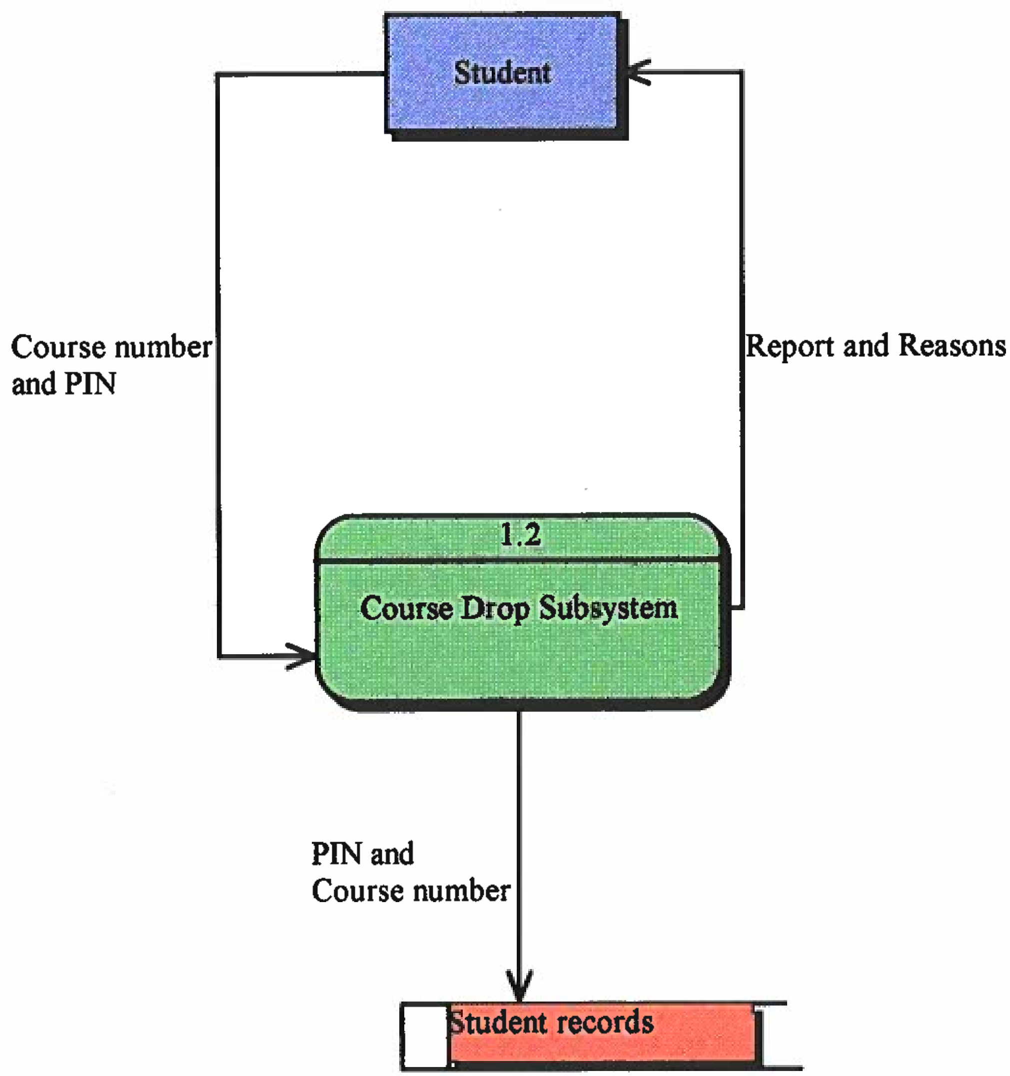


DFD for Course Addition Subsystem
Level 1

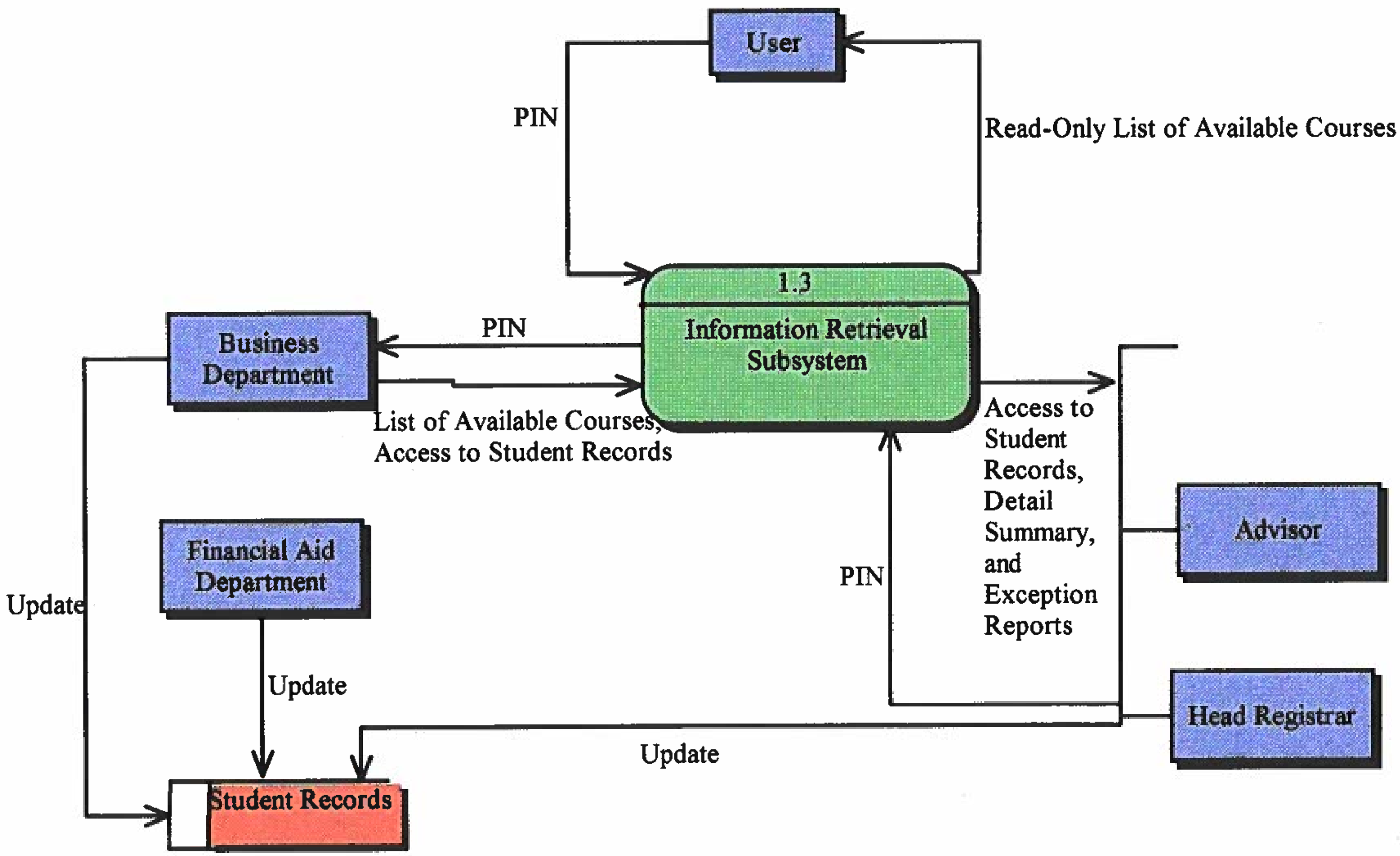


Go to page 12 for explosion

DFD for Course Drop Subsystem
Level 1

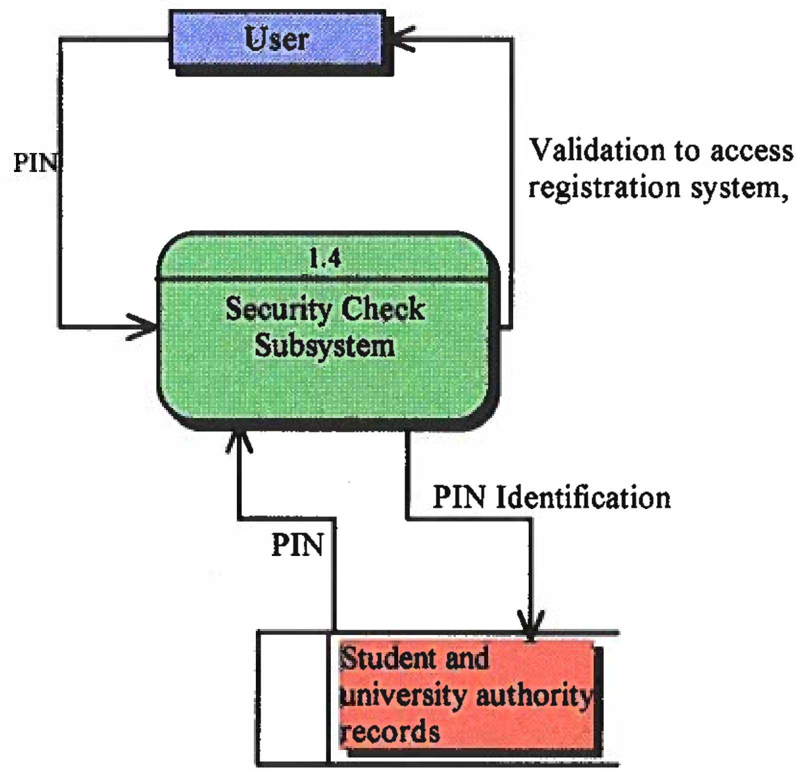


DFD for Information Retrieval Subsystem
Level 1



Go to page 14 for explosion

DFD for Security Check Subsystem
Level 1



MILESTONE 5

SUBMISSION CHECKLIST AND COVER SHEET

DATE: October 13, 1997

TO : Dr. Linda Behrens

FROM: TEAM MEMBERS WASHINGTON, AGELA
RAGHURAMAN, ANANTH

We are submitting the following for your evaluation.

Milestone 5 checklist:

[x] We are resubmitting our System Profile (Milestone 2).

[x] We are submitting a decomposition diagram for review.

[x] We are submitting a leveled set of data flow diagrams for review.

[x] Our DFDs are complemented by brief written narratives (one page or less per diagram).

[ ] We are submitting the following additional documentation:

____

____

We have carefully checked our data flow for the following:

[x] There are no black holes or miracles.

[x] We have used good, descriptive names throughout the decomposition diagram and data flow diagrams

[x] Our data flow diagrams are balanced.

[x] With the exception of the context diagram, every DFD has more then one process.

[ ] Special comments to the instructor:______________________

____

To be completed by the instructor:

I have reviewed Milestone 5. My evaluation follows:

[ ] Milestone returned ungraded because of__________________

____

[ ] Milestone acceptable as submitted. Grade of 30 +5 for legends recorded in students' record.

[ ] Milestone acceptable as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

[ ] Milestone **NOT** acceptable as submitted. Grade of__________ recorded in students' record.

[ ] Milestone **NOT** acceptable as submitted because_____________

____

Resubmit by date of___________ to establish a permanent grade.

SYSTEM PROFILE

BUSINESS NAME: CHARLESTON SOUTHEN UNIVERSITY

INDUSTRY TYPE: EDUCATION

FUNCTIONAL BUSINESS SYSTEM: THE MANUAL REGISTRATION PROCESS

1) Problems and opportunities that triggered this project:

In the current registration process, students go to their respective advisors, with a list of courses they want, to get their (the advisors') approval. Each advisor then goes through the list to see if the student has the necessary prerequisites. If the student doesn't the advisor asks him/her to take another course mix. Then, the student goes to the registrar's office and submits the list. The registrar goes over the list and singles out the courses that have been filled up. These, the student cannot take. The student then has to repeat the process.

Problems with the current system:

A) The student has to go to the registrar's office to register. If it is impossible for the student to do so, for any reason, he/she cannot register.

B) The student has to repeatedly visit his/her advisor if he/she wants to drop/add a course, to get approval for the new addition.

C) The registrar has to manually check for available courses and this is tedious. Even as this process goes on, the other courses may be filled up and the students might end up making another trip to his advisor.

Charleston Southern University has, for the past few years, been making an effort to raise enrollment, and the registration system is one of its business functions it wants to upgrade to provide better service.

Cont...

2) Purpose of the business application, its business goals and the information system objectives directed toward these goals:

Business Application's Purpose: The purpose of the new system is to facilitate the process of registration. It does so by reducing the physical exertion involved in the task of registration.

Business Application's Goals:

- To allow students to register for courses remotely and by themselves, for the most part.
- To reduce the pressure on the advisor, by reducing the number of students who want counseling.
- To reduce the pressure on the registrars.
- To reduce errors, during the process. For example, errors like registering without prerequisites or the registrar registering the student for the wrong course.

Information System's objectives:

- Improved business system performance:
  To make the registration system faster and efficient, so that the student doesn't have to stand in a queue and to update student records as soon as possible so available courses are made known continuously.
- Improved decision making:
  Since the student knows which courses are available, as soon as possible, he/she can immediately decide to add/drop a particular course.
- Lower business costs:
  Since registrars don't have to register the total student population, the university can employ minimum number of personnel; so there is a reduction in total overtime pay.

Cont...

- Improved business control:
  This objective aims at better control of university authorities (professors and management etc.) over the registration process. University authorities would be able to check on the status of any student without tedious effort. Only the registrar and authorized personnel would be able to access records. These are the security features.
- Fewer errors:
  The system aims at reducing human errors during the process.
- Better service and morale:
  Reducing long queues and errors would make a big impression on the student. Employees would also have more time and energy to devote to other routine tasks.

System policies that constrain the information system solution:

- The system must obey Federal Privacy Laws, and the Privacy Laws of the university. In simple terms, these laws rule that, without the consent of the student his/her registration information cannot be disclosed to other people.
- The system shouldn't allow the student to register courses which are taught the same time.

Cont...

3) The System's user community and their roles:

Direct Users

- The direct users are students who register for their courses.
- The other direct users are the registrar representatives who register students, who for some reason, come by to register manually.

Managers including: supervisors, middle management, and executive management who will receive information from the system and/or make decisions based on information from the system:

- Rex Nestor, Head Registrar: He uses the system to prepare detail, exception and summary reports.

Indirect Users

Other departments, offices, or divisions within the business who, while not directly using the system, may be affected by the system or receive information from it:

- Indirect users are, the accounting department, business administration office and the Dean of Students.

People or organizations external to the business which will or may be affected by the system.

- Federal and State Financial Aid Depts. and other Scholastic/Philanthropic organizations which require a student's registration information to process his/her application for scholarship/financial aid.

4) Capabilities and support the system will provide. Information system capabilities can be categorized as inputs, outputs, and decision making opportunities.

Input transactions to be processed:

- Course addition: During the registration process, each student adds his/her courses.
- Course drop: A student can drop courses.
- The Head Registrar can update/view student records.

Cont…

- Hold: The Registrar can put a hold on a student's registration.
- Queries for available courses.

Output transactions to be generated:

- Retreival of registration information by students.
- Answering questions about available courses.
- Information retrieval on any student and the facility to update it by the Registrar.

Outputs or Information/Reports to be generated (include a statement of purpose):

Detail Reports:

- Since departments such as Financial Aid want to know all the details of a student's registration, the system generates detailed reports. These reports are also generated as backup, in the event of any mishap involving the hard drive or floppy disks.

Exception reports:

- Exception reports provide for retrieving information on students who have been put on hold, and indicate reason(s).

Summary reports:

- The system provides summary reports so the Head Registrar can report statistical trends in the student population. For e.g., average GPA, minimum GPA and maximum GPA.

Decision support:

- When the student wants to take a course, the system tells him/her if it is available or not.
- If the student, for any reason, wants to register manually, the system tells the registrar whether the student has the required prerequisites.

Cont...

5) All business systems use networks, that is, the distribution structure of people, data, activities, and technology to business locations and the movement of those building blocks between those locations. Remember, each building, each floor, each office may be a different business location. Describe the business network.

Locations where business activity is performed:

- o The registration network consists of the Head Registrar, assistant registrar, representatives, and students. Other people might access the network just to get general information about available courses.
- o Students register for their courses, including dropping, at any remote location, via a touch-tone phone. They dial (803) 863 EXT. and a voice prompt asks them to select from several options (a flow chart of the prototype is enclosed). They can also retrieve other accessible information from that site.
- o Registrars retrieve printed information, transcripts and status checking at the registrar's office.

6) Current technology already supports the other four building blocks of the existing business system. You may design new technology to support various aspects of the new information system later in the project. What hardware and software is being used to capture, store and manage data? What hardware and software is required to support the information system activities described earlier? What communications technology is used to interconnect data and process technology at different business locations described earlier?

Computer Hardware to be used (if known):

A local area network and communications hardware like modems.

Computer software to be used (if known):

An AT&T communication software package will be used.

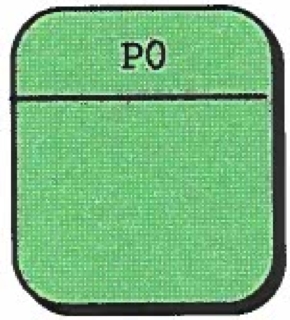
Reader's guide and legend

External Agent:	A person or Organizational unit which lies outside the system, but interacts with it. The External Agents in the Registration System are: Student, Advisor, Head Registrar, Assistant Registrars, Business Department and Financial Aid Department.
Data Store:	A Data Store is a collection of data. The Data Stores in the registration system are: Course List, Available Course List, Prerequisite Course List and Student Records.
Data Flow:	It is the input of data to a process. For example, in the registration system, the student inputs the PIN, which is a flowing data, into the process Course Addition Subsystem.
Processes:	Processes which operate on data to get work done or to turn it into useful information. In the registration system general processes are: Course Addition Subsystem (the data for this is the course number and PIN), Course Drop Subsystem, Information Retrieval Subsystem, and Security Check Subsystem.

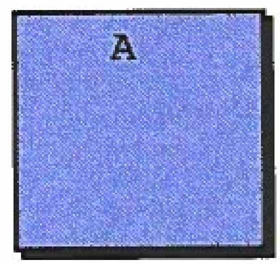
Automatic Inc.

Legend

Processes........................

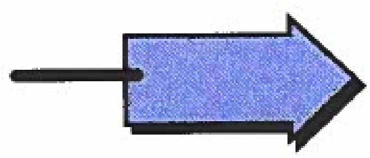


Agent........................



Data Stores.....................

Data Flows....................



MILESTONE 6

SUBMISSION CHECKLIST AND COVER SHEET

DATE: October 20, 1997

TO: Dr. Linda Behrens

From: TEAM MEMBERS: WASHINGTON, ANGELA
RAGHURAMAN, ANANTH

We are resubmitting the following milestone for your evaluation.

Milestone 6 Checklist:

[X] We are resubmitting our System Profile (Milestone 2).

[X] We are submitting a set of location connectivity diagrams for review.

[X] We are submitting a LCD legend and reader's guide.

[X] Our LCDs are complemented by brief written narratives (one page or less per diagram).

[ ] We submitting the following additional documentation:

____

We have carefully checked our location connectivity diagram for the following:

[X] We have used good, descriptive names throughout the location connectivity diagrams.

[X] Our location connectivity diagrams are balanced.

[ ] Special comments to the instructor: ______________

____

To be completed by the instructor:

I have reviewed Milestone 6. My evaluation follows:

[ ] Milestone returned ungraded because of ______________

____

[ ] Milestone acceptable as submitted. Grade of 40 ____________
recorded in students' record

[ ] Milestone acceptable as submitted; however, it needs to be corrected and resubmitted to establish a permanent grade.

[ ] Milestone **NOT** acceptable as submitted. Grade of __________ recorded in students' record.

[ ] Milestone **NOT** acceptable as submitted because _____________

____

Resubmit by date of __________________ to establish Permanent grade.

SYSTEM PROFILE

BUSINESS NAME: CHARLESTON SOUTHEN UNIVERSITY

INDUSTRY TYPE: EDUCATION

FUNCTIONAL BUSINESS SYSTEM: THE MANUAL REGISTRATION PROCESS

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Cont...

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- The other direct users are the registrar representatives who register students, who for some reason, come by to register manually.

Managers including: supervisors, middle management, and executive management who will receive information from the system and/or make decisions based on information from the system:

- Rex Nestor, Head Registrar: He uses the system to prepare detail, exception and summary reports.

Indirect Users

Other departments, offices, or divisions within the business who, while not directly using the system, may be affected by the system or receive information from it:

- Indirect users are, the accounting department, business administration office and the Dean of Students.

People or organizations external to the business which will or may be affected by the system.

- Federal and State Financial Aid Depts. and other Scholastic/Philanthropic organizations which require a student's registration information to process his/her application for scholarship/financial aid.

4) Capabilities and support the system will provide. Information system capabilities can be categorized as inputs, outputs, and decision making opportunities.

Input transactions to be processed:

- Course addition: During the registration process, each student adds his/her courses.
- Course drop: A student can drop courses.
- The Head Registrar can update/view student records.

Cont…

- Improved business control:
  This objective aims at better control of university authorities (professors and management etc.) over the registration process. University authorities would be able to check on the status of any student without tedious effort. Only the registrar and authorized personnel would be able to access records. These are the security features.
- Fewer errors:
  The system aims at reducing human errors during the process.
- Better service and morale:
  Reducing long queues and errors would make a big impression on the student. Employees would also have more time and energy to devote to other routine tasks.

System policies that constrain the information system solution:

- The system must obey Federal Privacy Laws, and the Privacy Laws of the university. In simple terms, these laws rule that, without the consent of the student his/her registration information cannot be disclosed to other people.
- The system shouldn't allow the student to register courses which are taught the same time.

Cont...

- Hold: The Registrar can put a hold on a student's registration.
- Queries for available courses.

Output transactions to be generated:

- Retreival of registration information by students.
- Answering questions about available courses.
- Information retrieval on any student and the facility to update it by the Registrar.

Outputs or Information/Reports to be generated (include a statement of purpose):

Detail Reports:

- Since departments such as Financial Aid want to know all the details of a student's registration, the system generates detailed reports. These reports are also generated as backup, in the event of any mishap involving the hard drive or floppy disks.

Exception reports:

- Exception reports provide for retrieving information on students who have been put on hold, and indicate reason(s).

Summary reports:

- The system provides summary reports so the Head Registrar can report statistical trends in the student population. For e.g., average GPA, minimum GPA and maximum GPA.

Decision support:

- When the student wants to take a course, the system tells him/her if it is available or not.
- If the student, for any reason, wants to register manually, the system tells the registrar whether the student has the required prerequisites.

Cont...

5) All business systems use networks, that is, the distribution structure of people, data, activities, and technology to business locations and the movement of those building blocks between those locations. Remember, each building, each floor, each office may be a different business location. Describe the business network.

Locations where business activity is performed:

- o The registration network consists of the Head Registrar, assistant registrar, representatives, and students. Other people might access the network just to get general information about available courses.
- o Students register for their courses, including dropping, at any remote location, via a touch-tone phone. They dial (803) 863 EXT. and a voice prompt asks them to select from several options (a flow chart of the prototype is enclosed). They can also retrieve other accessible information from that site.
- o Registrars retrieve printed information, transcripts and status checking at the registrar's office.

6) Current technology already supports the other four building blocks of the existing business system. You may design new technology to support various aspects of the new information system later in the project. What hardware and software is being used to capture, store and manage data? What hardware and software is required to support the information system activities described earlier? What communications technology is used to interconnect data and process technology at different business locations described earlier?

Computer Hardware to be used (if known):

A local area network and communications hardware like modems.

Computer software to be used (if known):

An AT&T communication software package will be used.

LCD Legend

Represents a moving and external location

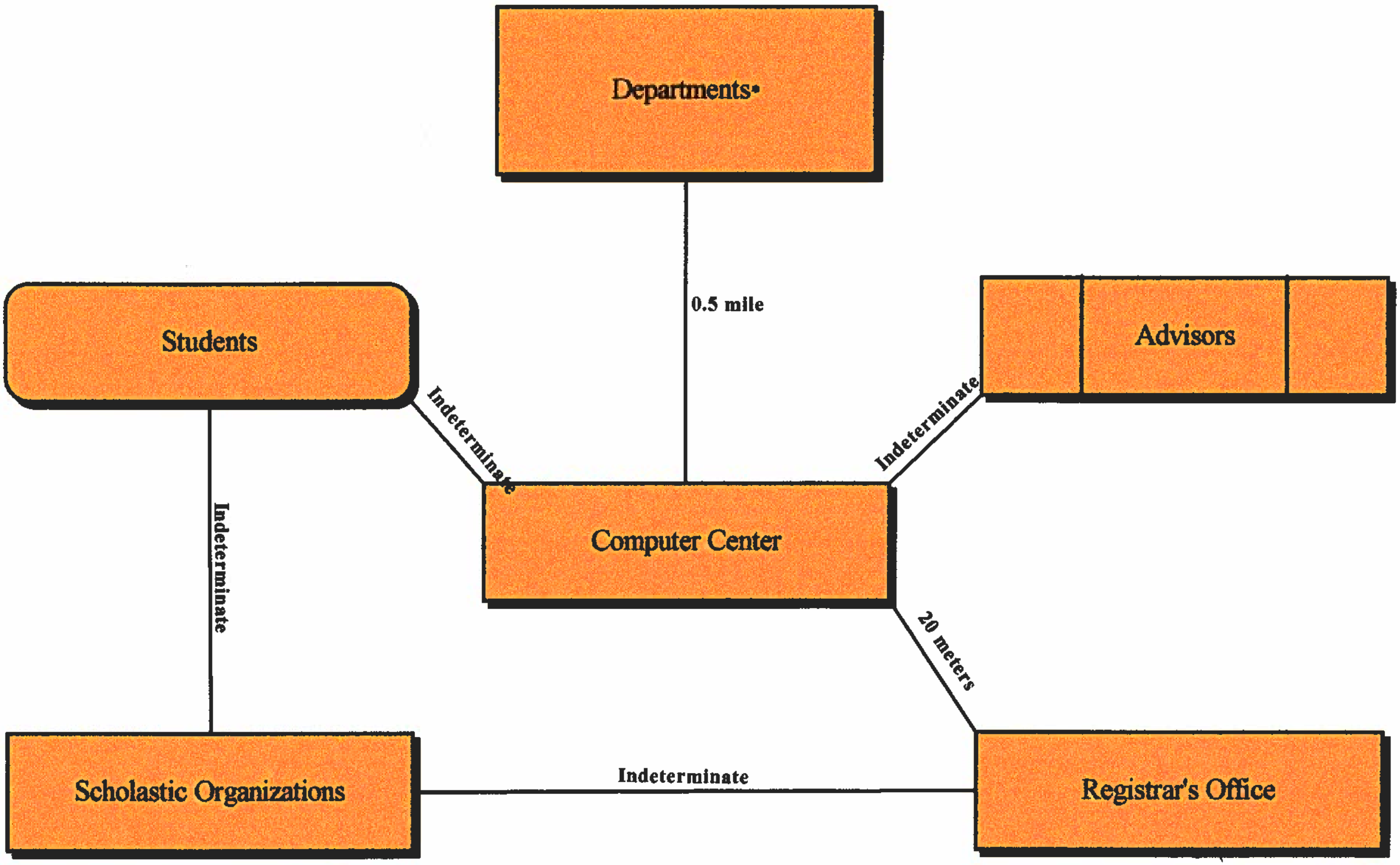
Represents a stationary location

A cluster of like locations

Length of connection

Connecting lines representing data transfer between locations

Location Connectivity Diagram
for CSU's Registration System



Brief written narrative

With regard to our project, redesigning CSU's registration system, the Location Connectivity Diagram (LCD) is just one level thick. This is because the users and hardware/software are only in two locations: either on-campus or off-campus. The locations represented in the LCD are, **Departments**: this includes the Business department and Financial Aid department, **Scholastic Organizations**, **Registrar's Office**, **Advisors**, **Students** and **Computer Center**. Departments are connected to **the Computer Center** because they need to get data from the computers inside the **Computer Center**. **Students** are connected to **the Computer Center** because they have to register or add/drop courses with the computers in the **Computer Center**. **Scholastic Organizations** aren't connected directly to the **Computer Center** because they are external organizations without access to the registration system. Instead, they are connected to the Students and Registrar's Office. There is data transfer between all the three of them. **Students** are represented as moving and external locations because they can register or add/drop from anyplace remotely; they may be off-campus, on-campus or on-the-move. Since the place for advisement changes from time to time and there are many **Advisors**, their locations are 'clustered' together and their distance from the **Computer Center** is indeterminate. A more detailed explanation is provided in the reader's guide.

Reader's guide

The main users of the registration system are the Students, Head Registrar, Assistant Registrars, Financial Aid and Business Departments, Advisors and any Scholastic Organizations. Then there is the Computer Center where a small network of computers are situated; this is the implementation location - the location where the actual process of registration and add/drop appear to go on. The logical locations - the places where the processes of registration/add/drop/information retrieval/updating go on - are the locations other than the Computer Center. If the students need to register or add/drop, this they can do remotely from where they are. The registration system in turn gives them reports on whether their attempt was successful. This is depicted by the line joining **Students** and **Computer Center**. The length of connection is indeterminate because students access the system from anyplace. The next locations are that of the **Departments**. The Financial Aid and Business departments need to know a student's registration information so that they can charge tuition, fees and room and board and determine the amount of financial aid. For this, they must have access to the registration system. Since all the student records are computerized and located in the Computer Center, they are connected to it. The **Registrar's Office** needs to be connected to the Computer Center because it might need to manually register some students. Also it might need to access student records to change, update or place holds.